

RESEARCH ARTICLE

Compressive Signal Processing for Estimating Range-Velocity-AoA in FMCW Radar Applications

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ABSTRACT Frequency-modulated continuous wave (FMCW) radars are known to accurately estimate the parameters of targets with low-cost and low-power transceiver systems. This work shows that some features associated with the spectrum of estimated parameters allow one to integrate the compressive sampling (CS) theory into the FMCW signal processing. To this end, we establish a new analytical framework through a tensor format to facilitate a systematic and convenient FMCW signal processing model. By observing the sparsity feature in the tensor of FMCW spectrum, we develop an extensive theoretical analysis to justify the use of CS theory in the FMCW radars, and it enabled us to propose a novel scheme, namely compressive FMCW signal processing, for estimating range, velocity, and angle-of-arrival (AoA) of the targets. The addition of CS theory allows our proposed scheme to use sampling-rates below Nyquist criterion, thus minimizing the number of sampled data and mitigating the issues related to high-rate and power-hungry analog-to-digital converter (ADC) in extremely high-frequency radar applications. Furthermore, the proposed compressive FMCW signal processing also significantly reduces the number of radio-frequency (RF) front-end elements necessary for estimating AoA, leading to a further saving in cost and power consumption of the FMCW radars. Despite having sub-Nyquist sampling-rates and reduced RF front-end elements, the performance evaluations show that our CS-based approach maintains the estimation capability and accuracy of conventional FMCW signal processing.

INDEX TERMS FMCW radars, compressive sampling, range, velocity, angle-of-arrival, parameters estimation.

I. INTRODUCTION

The frequency-modulated continuous wave (FMCW) radars have become popular tools to detect targets and estimate their parameters due to several key benefits, such as low-complexity radar transceiver [1], low-power requirement for signal transmission [2], and highly accurate measurement of parameters [3]. Typically, the estimated parameters are range and velocity of the targets from the radar. The features of FMCW radar make it attractive for various applications such as LiDAR system [4], automotive radars [5], synthetic aperture radar (SAR) [6], weather radar [7], through-the-wall radar [8], and maritime radar [9].

FMCW radar transmits a continuous sinusoid wave signal called chirp, whose frequency linearly increases over time, and the receiver intercepts the reflected chirp signal from the targets. Since the reflected chirp signal is a time-delayed version of the transmitted one, a frequency difference (or beat frequency) between those two chirp signals is exploited to estimate the targets' parameters. The beat frequency is conveniently measured by mixing the two chirp signals and passing the mixer output through a low-pass filter. It is shown in [3] that the beat frequency is proportional to the distance of targets from the receiving antenna, thus enabling range measurement of the targets. In addition, the velocity of moving targets can also be estimated by transmitting a signal containing several equi-spaced chirps with a linear sawtooth frequency modulation [3], [10], [11]. These chirps have a specific duration, and they are sent in one FMCW

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frame. Due to the Doppler effect, the beat frequencies vary across different chirp durations in the received FMCW frame, and this phenomenon is exploited to compute the velocity of moving targets. In the FMCW radar receiver, the mixing and filtering processes of the received FMCW frame are performed by a radio-frequency (RF) front-end element, which outputs an intermediate frequency (IF) signal. In [12] and [13], two-dimensional fast Fourier transform (2D-FFT) is introduced as a signal processing tool to estimate range and velocity from the IF signal. The first FFT operation, labeled as range-FFT, is applied to the time-domain samples of each chirp signal. The second FFT calculation, called Doppler-FFT, is then performed on the output of the range-FFT over the chirp domain, i.e., different chirps in a single FMCW frame. The 2D-FFT computation results in a 2D range-Doppler-frequency spectrum-map that is further processed to produce range and velocity information [12], [13].

In addition to range and velocity, another useful parameter to be measured is the direction of the targets relative to the receiver. In [14], a Wellen radar (WERA) is used to determine ocean surface currents, waves, and the direction of the wind. Specifically, WERA measures the azimuth angle of the ocean surface currents by using angle-of-arrival (AoA) estimation. In [15], Bartlett beamforming and multiple signal classification (MUSIC) are utilized to evaluate the surface currents in high-frequency surface wave radar (HFSWR) system. Furthermore, [16] focuses on the range-azimuth map and develops a novel algorithm based on the 2D-MUSIC method in order to simultaneously locate the range and the azimuth of the targets in the FMCW radar systems. For estimating the horizontal direction of the targets, namely the azimuth angle or AoA, an array of receiving antennas is required. The differential distance from the targets to each antenna in the array is exploited to calculate the AoA of the targets with respect to the array [17]. Measuring the AoA of targets necessitates additional signal processing for the received signals at spatial domain, i.e., signals obtained from multiple receiving antennas, and the number of these antennas controls the sensitivity of the AoA estimates. However, adding more physical antennas entails extra RF front-end processing elements, thus prohibiting the FMCW radar in low-cost applications.

Prior to data processing in the digital signal processor (DSP), the analog IF signal needs to be represented in its digital samples by an analog-to-digital converter (ADC), and the sampling-rate used in the ADC is at least Nyquist-rate to avoid aliasing [18]. It is shown in [3] that the number of samples in one chirp determines the maximum coverage distance of FMCW radar systems, and a large amount of digital samples is often required to estimate the targets at a longer range. However, obtaining a high number of data samples can be translated into having power-hungry sampling since the power consumption of ADC is approximately proportional to its sampling-rate [19]. In addition, large data transfer to the DSP increases the complexity of signal

processing units in FMCW radar receivers. The issues associated with the high number of data samples are even more problematic in extremely high-frequency radar applications, such as mmWave radars, which necessitate a very high sampling-rate in the ADC.

Aiming to reduce the sampling-rate in the ADC, a theory of compressive sampling (CS) is introduced in [20]. It is shown in [21] that the reconstruction of an analog signal from digital samples produced by the sampling process with sub-Nyquist rate is possible (with no or minimum error) if the analog signal possesses a sparse structure or it contains a few significant values in a particular basis. The CS problem can be viewed as an underdetermined system of linear equations in which the observed compressive measurements are connected to the sparse vectors (solutions to be recovered) via a multiplication between the dictionary matrix and the measurement matrix. Since the product of these two matrices results in a fat matrix, some mathematical conditions, such as mutual coherence and restricted isometry property (RIP) [20], [21], corresponding to the resulting system matrix are necessary for a successful reconstruction of the sparse vector. Several algorithms to find a sparse solution of underdetermined systems have been proposed, such as orthogonal matching pursuit (OMP) [22], least absolute shrinkage, and absolute operator (LASSO) [23], [24], and focal underdetermined system solver (FOCUSS) [25].

The addition of CS theory in radar applications has been discussed in [7], [26], [27], [28], [29], and [30]. In [7], the CS theory is utilized to reduce polarimetric Doppler weather radar data by exploiting the sparsity of the beat signal in the transform domain. The CS-based approach with coordinate descent LASSO (CD-LASSO) is used in [26] for estimating the AoA of the targets, and [27] extends this work by including range estimation using a similar CS-based technique. The work of [28] uses the CS theory with OMP algorithm in FMCW-SAR applications, and it aims to reconstruct a sparse radar scene from the received beat signals. Furthermore, the scaled CS-based method is introduced in [29] for reconstructing sparse radar scenes in automobile applications. In [30], the CS-based method using Newton smooth ℓ_0 -norm (NSLO) algorithm is incorporated to obtain 3D images based on the received THz FMCW-SAR signals.

In this paper, we propose a compressive FMCW signal processing in which the CS theory is utilized in the FMCW radar receiver for estimating range, velocity, and AoA of the targets. The CS theory is implemented by exploiting the sparsity feature of the 3D range-Doppler-AoA spectrum-map obtained from signal processing, since the 3D spectrum-map contains mostly zeros or insignificant values when the number of detected targets is considerably less than the number of grid-points in the map. To the best of our knowledge, this work is the first one to introduce three domains compression tailored for FMCW radar applications, i.e., time, chirp, and spatial domain compressions, since [7],

[26], [27], [28], [29], [30] only implement CS in less than three domains with different estimated parameters. In particular, our proposed compressive FMCW signal processing introduces spatial domain compression to reduce the number of RF front-end elements in the FMCW receiver. Subsequently, the time and chirp domain compressions are applied to the signals produced by the RF front-end elements, which leads to a large amount of reduction in the resulting digital samples. By using the compressed samples, the reconstruction is performed to obtain the 3D range-Doppler-AoA spectrum-map. This work will show that our CS-based approach has minimal impact on the capability and accuracy of original FMCW radars in estimating range, velocity, and AoA through numerical simulations. As a result, the FMCW radar receivers employing our proposed compressive signal processing require less space complexity, lower power consumption, and cheaper implementation cost, compared to the conventional FMCW signal processing.

The remainder of this paper is organized as follows. In Section II, some basic notations and mathematical models for the FMCW signals are presented. Subsequently, we present the continuous and discrete signal processing framework for the FMCW radar to estimate the range, the velocity, and the AoA of the targets in Section III. In Section IV, our proposed compressive FMCW signal processing to estimate range, velocity, and AoA is discussed. We also derive the analytical expressions of the compression processes and present the theoretical results for justifying the reconstruction processes. In Section V, some numerical results for evaluating the estimated parameters are presented. Finally, Section VI concludes this work.

II. PRELIMINARIES

A. BASIC DEFINITIONS AND NOTATIONS

In this paper, vectors are assumed to be column vectors, and a $N \times 1$ vector is written as $\mathbf{x} = [x_0, x_1, \dots, x_N]^T$ where T denotes the transpose operation. The corresponding $1 \times N$ row vector is expressed as $\mathbf{x}^T = [x_0, x_1, \dots, x_N]$. Under some clear contexts, column vectors and row vectors are simply called columns and rows, respectively. $M \times N$ matrix $\mathbf{X}$ can be represented in terms of their columns or rows, i.e.,

$$\mathbf{X} = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N] = [\tilde{\mathbf{x}}_0, \tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_M]^T,$$

where $\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_N$ and $\tilde{\mathbf{x}}_0, \tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_M$ are the columns and rows of $\mathbf{X}$, respectively. In addition, this work will incorporate a special type of square matrix, i.e., the discrete Fourier transform (DFT) matrix, represented as $\mathbf{F}_S$ where S specifies its size, and the corresponding inverse DFT matrix is given by $\mathbf{F}_S^{-1}$.

A continuous-time signal is denoted as $x(t)$, while its discrete version is expressed as $x[n]$. A collection of L continuous-time and discrete-time signals is represented as a vector, i.e., $\mathbf{x}(t) = [x_0(t), x_1(t), \dots, x_{L-1}(t)]^T$ and $\mathbf{x}[n] = [x_0[n], x_1[n], \dots, x_{L-1}[n]]^T$, respectively. Furthermore, the vectors containing a collection of signals are stored in a

matrix, e.g., a matrix of $L_1 \times L_2$ continuous-time signals $\mathbf{X}(t)$ is given by

$$\mathbf{X}(t) = [\mathbf{x}_0(t), \mathbf{x}_1(t), \dots, \mathbf{x}_{L_2-1}(t)],$$

where $\mathbf{x}_i(t) = [x_{i,0}(t), x_{i,1}(t), \dots, x_{i,L_1-1}(t)]^T$ is the $(i+1)$ -th column of $\mathbf{X}(t)$, and a matrix of $L_1 \times L_2$ discrete signals $\mathbf{X}[n]$ is expressed as

$$\mathbf{X}[n] = [\mathbf{x}_0[n], \mathbf{x}_1[n], \dots, \mathbf{x}_{L_2-1}[n]],$$

where $\mathbf{x}_i[n] = [x_{i,0}[n], x_{i,1}[n], \dots, x_{i,L_1-1}[n]]^T$ is the $(i+1)$ -th column of $\mathbf{X}[n]$. In the later part of our paper, the discrete-time signal processing is presented through a tensor format,* and we follow the conventions introduced in [31] for the algebraic of tensor. A complex-valued tensor having a size of $I \times J \times K$ is written as $\mathcal{X} \in \mathbb{C}^{I \times J \times K}$. Accessing a matrix slice in $\mathcal{X}$ is performed with either $\mathbf{X}(i, :, :)$, $\mathbf{X}(:, j, :)$, or $\mathbf{X}(:, :, k)$. A matrix slice can be represented by its columns or rows, e.g. $\mathbf{X}(:, :, k)$ is expressed as

$$\begin{aligned} \mathbf{X}(:, :, k) &= [\mathbf{x}(:, 0, k), \mathbf{x}(:, 1, k), \dots, \mathbf{x}(:, J-1, k)] \\ &= [\tilde{\mathbf{x}}(0, :, k), \tilde{\mathbf{x}}(1, :, k), \dots, \tilde{\mathbf{x}}(I-1, :, k)]^T, \end{aligned}$$

where $\mathbf{x}(:, j, k)$ for $j = 0, 1, \dots, J-1$ and $\tilde{\mathbf{x}}(i, :, k)$ for $i = 0, 1, \dots, I-1$ are the columns and rows of $\mathbf{X}(:, :, k)$, respectively.

B. FMCW SIGNAL MODEL

In this subsection, we briefly introduce the mathematical model of a FMCW system [32], [33], [34], [35], [36]. Let us consider the transmitted chirp signal $s_0(t)$ which is given by

$$s_0(t) = \begin{cases} V_s \cos \left\{ 2\pi \left(f_c t + \frac{\alpha t^2}{2} \right) \right\}, & \text{for } 0 \leq t < T_c, \\ 0, & \text{elsewhere,} \end{cases} \quad (1)$$

where V_s is the amplitude of $s_0(t)$, f_c is the carrier frequency, α is the slope of the FMCW chirp (the rate of frequency change over time), and T_c is the duration of the chirp. This work considers P moving targets in the far field region of the receiving antennas, and $s_0(t)$ is reflected to the FMCW receiver by these P objects. Let us denote the reflection or the back-scatters coming from the p -th target as $b_{p,0}(t)$ for $p = 1, 2, \dots, P$, and it is expressed as

$$b_{p,0}(t) = V_p \cos \left\{ 2\pi \left(f_c(t - \tau_p(t)) + \frac{\alpha}{2}(t - \tau_p(t))^2 \right) \right\}, \quad (2)$$

for $0 \leq t - \tau_p(t) < T_c$, and $b_{p,0}(t) = 0$, elsewhere, V_p is the amplitude of $b_{p,0}(t)$, and $\tau_p(t)$ is the time delay of the received reflected chirp from the p -th object. We also denote R_p as the distance of the p -th object relative to the FMCW receiver. The relation between $\tau_p(t)$ and R_p is given by

$$\tau_p(t) = 2\{R_p + v_p t\}/c, \quad \text{for } p = 1, 2, \dots, P, \quad (3)$$

*Without loss of generality, we simply call a three-dimensional array as a tensor throughout this paper.

where v_p is the relative velocity of the p -th object in the direction towards the FMCW receiver, and c is the speed of light in the vacuum. Since $V_p \ll V_s$ in general, $b_{p,0}(t)$ is a delayed and attenuated version of $s_0(t)$.

For detecting velocity of targets, the chirp waveform in (1) is repeatedly transmitted within a FMCW frame. The number of chirps and the duration of a FMCW frame are denoted by N_c and T_f , respectively. Hence,

$$T_f = T_c N_c,$$

and, the chirp bandwidth B_c is given by

$$B_c = \alpha T_c. \quad (4)$$

As a result, the $(n_c + 1)$ -th transmitted chirp $s_{n_c}(t)$ is expressed as

$$s_{n_c}(t) = V_s \cos \left\{ 2\pi \left(f_c(t - n_c T_c) + \frac{\alpha}{2}(t - n_c T_c)^2 \right) \right\}, \quad (5)$$

for $n_c T_c \leq t < (n_c + 1)T_c$, and $s_{n_c}(t) = 0$, elsewhere, with $n_c = 0, 1, \dots, N_c - 1$. It is worth mentioning that (5) is a generalized version of (1), and $s_0(t)$ can be obtained by simply setting $n_c = 0$ in (5). Furthermore, the reflection of the $(n_c + 1)$ -th chirp coming from the p -th object $b_{p,n_c}(t)$ is given by

$$b_{p,n_c}(t) = V_p \cos \left\{ 2\pi \left(f_c[t - n_c T_c - \tau_p(t)] + \frac{\alpha}{2}[t - n_c T_c - \tau_p(t)]^2 \right) \right\}, \quad (6)$$

for $n_c T_c \leq t - \tau_p(t) < (n_c + 1)T_c$, and $b_{p,n_c}(t) = 0$, elsewhere. Similarly, we obtain $b_{p,0}(t)$ in (2), when putting $n_c = 0$ in (6). By using (5), the transmitted FMCW frame $s(t)$ is compactly written as

$$s(t) = \sum_{n_c=0}^{N_c-1} s_{n_c}(t) = V_s \cos \left\{ 2\pi \left(f_c(t \bmod T_c) + \frac{\alpha}{2}(t \bmod T_c)^2 \right) \right\}, \quad (7)$$

for $0 \leq t < N_c T_c$ and $t \bmod T_c$ represents the remainder of t/T_c . By considering (6), the reflection of the FMCW frame coming from the p -th target $b_p(t)$ is expressed as

$$\begin{aligned} b_p(t) &= \sum_{n_c=0}^{N_c-1} b_{p,n_c}(t) \\ &= V_p \cos \left\{ 2\pi \left(f_c([t - \tau_p(t)] \bmod T_c) + \frac{\alpha}{2}([t - \tau_p(t)] \bmod T_c)^2 \right) \right\}, \end{aligned} \quad (8)$$

for $p = 1, 2, \dots, P$ and $0 \leq t - \tau_p(t) < N_c T_c$.

The sensitivity of FMCW radar in distinguishing multiple targets with relatively similar ranges and velocities is controlled by the resolution of estimation. The range resolution ΔR_{res} depends on B_c , and it is given by $\Delta R_{\text{res}} = c/(2B_c)$ [3]. Furthermore, the velocity resolution Δv_{res} is controlled by T_c and N_c as $\Delta v_{\text{res}} = c/(2f_c T_f) = c/(2f_c T_c N_c)$ [37]. The

maximum range that can be estimated R_{max} is related to sampling rate N_s/T_c and α as [3]

$$R_{\text{max}} = \frac{cN_s/(2T_c)}{2\alpha} = \frac{cN_s/(2T_c)}{2(B_c/T_c)} = \frac{cN_s}{4B_c}, \quad (9)$$

where N_s is the number of samples in a chirp used by the discrete signal processing, and we will discuss it further in Section III. In addition, the maximum velocity that can be estimated is given by [37]

$$v_{\text{max}} = c/(4f_c T_c) = \Delta v_{\text{res}} N_c/2. \quad (10)$$

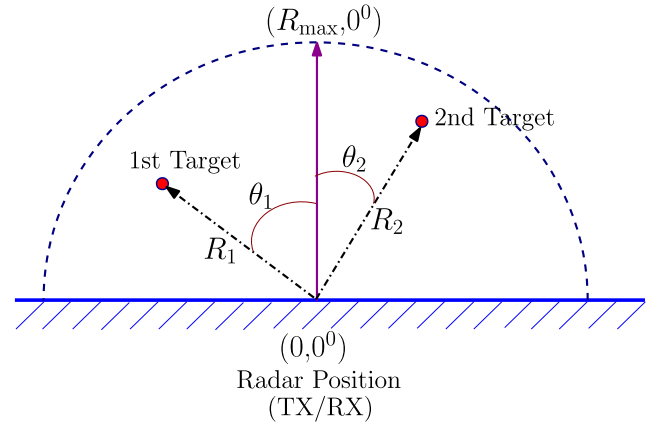


FIGURE 1. A geometric representation of two objects in front of FMCW system that is visualized on a flat surface with polar coordinate system.

Now, we consider a uniform linear array (ULA) of L omnidirectional elements at the FMCW receiver to estimate the AoA. In this work, we use assumptions based on [17] as follows:

- The objects are sufficiently far from the FMCW receiver, thus the reflected waves that arrive at the ULA are assumed as planar waves.
- The total bandwidth of the entire reflections received by the ULA does not exceed the inverse of the propagation delay from the first to the last antenna in the array, hence the delay observed between the elements in the array is regarded as a phase shift.

By using these assumptions, we introduce $x_l(t)$ as the output of the $(l + 1)$ -th antenna in the ULA, and it can be expressed as a combination of the aggregated reflections $b_p(t)$ in (8) for $p = 1, 2, \dots, P$, that is

$$x_l(t) = \sum_{p=1}^P a(\theta_p)^l b_p(t), \quad l = 0, 1, \dots, L - 1, \quad (11)$$

where θ_p is the AoA of the reflection coming from the p -th object $b_p(t)$ with respect to the ULA, $a(\theta_p)^l = \exp(j2\pi l \Delta \sin(\theta_p))$ is the response of the $(l + 1)$ -th antenna in the ULA to the object's reflection at the AoA of θ_p , and Δ is the distance between two consecutive antennas in the ULA in wavelength unit. This paper assumes $\Delta = 0.5$ to prevent spatial aliasing. The geometric representation of the FMCW system for detecting $P = 2$ targets is shown in Fig. 1. The distance from the FMCW receiver to the p -th target is denoted

as R_p , and Fig. 1 has $p \in \{1, 2\}$. The AoA of the p -th target is measured with respect to the line perpendicular to the array aperture, and θ_p indicates the AoA of the targets in Fig. 1.

III. SIGNAL PROCESSING MODEL FOR FMCW RADARS

This section discusses our signal processing framework for the aggregated FMCW signals from each antenna in the ULA represented in (11). First, we present the RF front-end processing of the received continuous-time FMCW signal and the sampling process necessary to obtain the discrete-time signal. Then, the data samples are expressed in a tensor format to facilitate systematic and convenient digital signal processing. The 3D-DFT operation is subsequently used for obtaining the 3D FMCW spectrum. Lastly, we describe unit transformations to extract the information of range, velocity, and AoA from the obtained 3D spectrum.

A. CONTINUOUS-TIME SIGNAL PROCESSING MODEL

In the FMCW receiver, the aggregated signals coming from multiple targets is first processed in the RF front-end connected to the $(l + 1)$ -th antenna. A mixer in the RF front-end produces a signal $\tilde{x}_l(t)$, i.e., the multiplication result between $x_l(t)$ in (11) and $s(t)$ in (7), which is given by

$$\tilde{x}_l(t) = x_l(t)s(t) = \sum_{p=1}^P a(\theta_p)^l b_p(t)s(t) \quad (12)$$

for $l = 0, 1, \dots, L - 1$. Then, $\tilde{x}_l(t)$ in (12) is passed through a low pass filter (LPF) having an impulse response of $h(t)$ to obtain the IF signal $x_{IF,l}(t)$, that is

$$\begin{aligned} x_{IF,l}(t) &= \tilde{x}_l(t) \star h(t) \\ &= \sum_{p=1}^P a(\theta_p)^l (b_p(t)s(t) \star h(t)) \\ &= \sum_{p=1}^P a(\theta_p)^l y_p(t) \end{aligned} \quad (13)$$

for $l = 0, 1, \dots, L - 1$, and

$$y_p(t) = b_p(t)s(t) \star h(t). \quad (14)$$

Now, let us consider (3) to compute the difference in the time delay $\Delta\tau_p$ within the chirp duration T_c as

$$\begin{aligned} \Delta\tau_p &= \tau_p(T_c) - \tau_p(0) \\ &= \frac{2\{R_p + v_p T_c\} - 2R_p}{c} = \frac{2v_p}{c} T_c, \end{aligned} \quad (15)$$

for $p = 1, 2, \dots, P$. Since generally $v_p \ll c$, we have $\Delta\tau_p \ll T_c$. By considering (6) and (15), the time difference between the beginning of the reflection of the two subsequent chirps coming from the p -th object is obtained as $T_c + \Delta\tau_p$. Since $\Delta\tau_p \ll T_c$, it is reasonable to have approximation

$$T_c + \Delta\tau_p \approx T_c. \quad (16)$$

By using (5), (6), and (16), we derive $b_p(t)s(t)$ in (14) as

$$b_p(t)s(t) = \sum_{n_c=0}^{N_c-1} b_{p,n_c}(t)s_{n_c}(t),$$

for $p = 1, 2, \dots, P$, and,

$$\begin{aligned} &b_{p,n_c}(t)s_{n_c}(t) \\ &= \left(\frac{V_p V_s}{2} \left[\cos \left(2\pi \{ 2f_c(t - n_c T_c) - f_c \tau_p(t) \right. \right. \right. \\ &\quad \left. \left. \left. + \alpha(t - n_c T_c)^2 + \frac{\alpha \tau_p(t)}{2} (\tau_p(t) - 2t + 2n_c T_c) \right\} \right) \right. \\ &\quad \left. \left. + \cos \left(2\pi \{ f_c \tau_p(t) - 0.5\alpha \tau_p(t)(\tau_p(t) - 2t \right. \right. \right. \right. \\ &\quad \left. \left. \left. + 2n_c T_c) \right\} \right) \right] \right) \end{aligned} \quad (17)$$

for $n_c T_c + \tau_p(n_c T_c) \leq t < (n_c + 1)T_c$, and $b_{p,n_c}(t)s_{n_c}(t) = 0$, elsewhere. Note that (17) indicates very high-frequency components around twice the carrier frequency f_c in $b_p(t)s(t)$. Hence, applying LPF on $b_p(t)s(t)$ removes the high-frequency components, and it allows us to express $y_p(t)$ in (14) as

$$y_p(t) = \sum_{n_c=0}^{N_c-1} y_{p,n_c}(t), \quad (18)$$

for $p = 1, 2, \dots, P$. Additionally, $y_{p,n_c}(t)$ is given by

$$\begin{aligned} y_{p,n_c}(t) &= V_y \cos \left(2\pi \{ f_{p,n_c}(t)(t - n_c T_c) \right. \\ &\quad \left. + f_c \tau_p(t) - \frac{\alpha \tau_p(t)^2}{2} \} \right) \end{aligned} \quad (19)$$

for $n_c T_c + \tau_p(n_c T_c) \leq t < (n_c + 1)T_c$, and $y_{p,n_c}(t) = 0$, elsewhere. It is worth mentioning that V_y in (19) is produced due to the filtering process, and $f_{p,n_c}(t)$ is the beat frequency corresponding to the p -th target and the $(n_c + 1)$ -th chirp. By using (3), $f_{p,n_c}(t)$ can be expressed as

$$f_{p,n_c}(t) = \alpha \tau_p(t) = 2\alpha(R_p + v_p t)/c, \quad (20)$$

for $p = 1, 2, \dots, P$, $n_c = 0, 1, \dots, N_c - 1$ and $n_c T_c + \tau_p(n_c T_c) \leq t < (n_c + 1)T_c$.

Due to $\Delta\tau_p \ll T_c$, we can approximate (20) to simplify our FMCW signal processing model, that is

$$\begin{aligned} f_{p,n_c}(t) &\approx f_{p,n_c} = \alpha \tau_p(n_c T_c) \\ &= 2\alpha\{R_p + v_p n_c T_c\}/c. \end{aligned} \quad (21)$$

Thus, (19) becomes

$$\begin{aligned} y_{p,n_c}(t) &\approx V_y \cos \left(2\pi \{ (t - n_c T_c) f_{p,n_c} \right. \\ &\quad \left. + f_c \tau_p(n_c T_c) - \frac{\alpha \tau_p(n_c T_c)^2}{2} \} \right), \end{aligned} \quad (22)$$

for $p = 1, 2, \dots, P$, $n_c = 0, 1, \dots, N_c - 1$, $n_c T_c + \tau_p(n_c T_c) \leq t < (n_c + 1)T_c$, and $y_{p,n_c}(t) = 0$, elsewhere. Note that $f_{p,n_c}(t)$ in (20) specifies the difference between the frequency of the transmitted chirp $s(t)$ and that of the chirp reflection $b_p(t)$ at

a specific time point, and it is proportional to the range of the p -th target from the FMCW receiver. In addition, (22) indicates that $y_p(t)$ in (18) is approximately a sequence of simple sinusoidal signals, each of which having a frequency given by f_{p,n_c} . Consequently, $x_{IF,l}(t)$ in (13) can be viewed as a linear combination of multiple sequences of simple sinusoidal signals, where each sequence is a contribution from the reflection of FMCW chirps coming from the p -th target.

The signal processing model for FMCW radar is demonstrated through a toy example shown in Fig. 2. In this example, a series of three chirps in a FMCW frame is transmitted, and we consider the existence of $P = 3$ targets located at different azimuth angle with respect to the receiving antennas. It is assumed that all targets are stationary or their movement leads to a negligible change in their range over the three chirps duration. An antenna of ULA receives the reflection of these chirps coming from $P = 3$ targets, and it is depicted by frequency-time plot in the RF stage of Fig. 2. After mixing and filtering processes, the IF signals for the $P = 3$ targets are generated, and they are shown in the IF stage of Fig. 2. Since the targets do not exhibit significant velocities, the IF signals have nearly the same frequency over the three-chirps duration, i.e., $f_{p,1} \approx f_{p,2} \approx f_{p,3} \approx f_p$. Fig. 2 clearly shows that the beat frequencies associated with multiple targets are different as they are proportional to the distance of the targets from the FMCW system. In subsection III-C, we discuss further signal processing techniques to utilize beat frequencies and multiple chirps to obtain the 2D range-velocity spectrum-map, which shows the range and velocity information of the three targets.

B. DISCRETE-TIME SIGNAL PROCESSING MODEL

Prior to signal processing in the DSP, the analog IF signal at the output of the $(l + 1)$ -th RF front-end element, i.e., $x_{IF,l}(t)$, needs to be sampled by the ADC. We denote the number of discrete samples of $x_{IF,l}(t)$ within one chirp duration as N_s , leading to a sampling-rate of N_s/T_c . After sampling, the ADC produces the discrete IF signal, i.e., $x_{IF,l,n_c}[n_s]$, which corresponds to the $(n_c + 1)$ -th chirp duration for $n_c = 0, 1, \dots, N_c - 1$ and $n_s = 0, 1, \dots, N_s - 1$. For a convenient mathematical representation, we express $x_{IF,l,n_c}[n_s]$ as $x_{IF,l}[n_s, n_c]$ and collect L discrete IF signals in a vector, that is

$$\mathbf{x}_{IF}[n_s, n_c] = [x_{IF,0}[n_s, n_c], \dots, x_{IF,L-1}[n_s, n_c]]^T, \quad (23)$$

for $n_s = 0, 1, \dots, N_s - 1$ and $n_c = 0, 1, \dots, N_c - 1$. In addition, we also represent $a(\theta_p)^l$ in a vector format as

$$\mathbf{a}(\theta_p) = [1, a(\theta_p), a(\theta_p)^2, \dots, a(\theta_p)^{L-1}]^T,$$

for $p = 1, 2, \dots, P$, and the $L \times P$ ULA response matrix corresponding to all P objects is neatly expressed as

$$\mathbf{A} = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_P)].$$

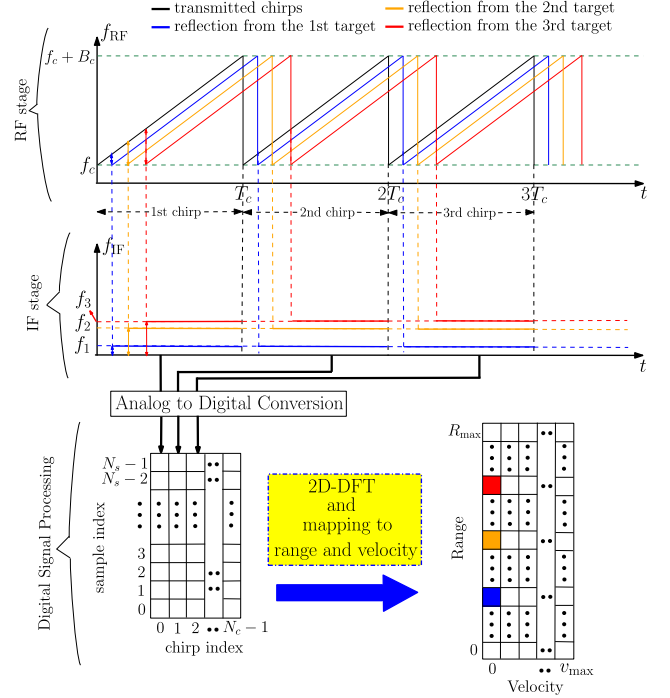


FIGURE 2. The transmitted FMCW frame containing three chirps, and the reception of these three chirps from $P = 3$ targets in one antenna of ULA. After a sampling process mentioned in Section III-B, the data samples are processed with 2D-DFT and unit transformation to obtain the corresponding 2D range-velocity spectrum-map. The 2D-DFT and unit transformation are discussed in Section III-C and III-D, respectively.

Therefore, by noting (13), $\mathbf{x}_{IF}[n_s, n_c]$ is defined as

$$\mathbf{x}_{IF}[n_s, n_c] = \sum_{p=1}^P \mathbf{a}(\theta_p) y_p[n_s, n_c] = \mathbf{A} \mathbf{y}[n_s, n_c], \quad (24)$$

where $y_p[n_s, n_c] = y_{p,n_c}[n_s]$ is the discrete version of $y_{p,n_c}(t)$ in (18), and

$$\mathbf{y}[n_s, n_c] = [y_1[n_s, n_c], y_2[n_s, n_c], \dots, y_P[n_s, n_c]]^T.$$

Next, we collect $\mathbf{x}_{IF}[n_s, n_c]$ for $n_s = 0, 1, \dots, N_s - 1$ into a $L \times N_s$ matrix, i.e.,

$$\mathbf{X}_{IF}[n_c] = [\mathbf{x}_{IF}[0, n_c], \mathbf{x}_{IF}[1, n_c], \dots, \mathbf{x}_{IF}[N_s - 1, n_c]].$$

As a result, we can express (24) in a more compact form as

$$\mathbf{X}_{IF}[n_c] = \mathbf{A} \mathbf{Y}[n_c], \quad (25)$$

where $\mathbf{Y}[n_c] = [\mathbf{y}[0, n_c], \mathbf{y}[1, n_c], \dots, \mathbf{y}[N_s - 1, n_c]]$ for $n_c = 0, 1, \dots, N_c - 1$.

C. 3D DISCRETE FOURIER TRANSFORM

In this subsection, we discuss the DFT operations on data samples stored in a tensor with respect to the time domain, chirp domain, and spatial domain. Representing data samples in a tensor provides us with a systematic and convenient approach to performing the digital signal processing, and it enables us to simultaneously obtain the information of range, velocity, and AoA of the targets from the processed data samples. Throughout this subsection, we utilize four

complex-valued tensors, i.e., $\mathcal{X}, \mathcal{Q}, \mathcal{R}, \mathcal{S} \in \mathbb{C}^{N_s \times N_c \times L}$ for storing data samples from signal processing.

First, we use (25) to fill $\mathcal{X}$, that is

$$\mathbf{X}(:, n_c, :) = \mathbf{X}_{\text{IF}}^T[n_c], \quad (26)$$

for $n_c = 0, 1, \dots, N_c - 1$, and (26) implies that $\mathcal{X}$ contains a collection of samples corresponding to various time indices, chirp durations, and receiving antennas. For exploiting the beat frequency in one chirp duration, we perform the DFT operation on each column of $\mathbf{X}(:, :, l)$, that is

$$\mathbf{Q}(:, :, l) = \mathbf{F}_{N_s} \mathbf{X}(:, :, l), \quad (27)$$

for $l = 0, 1, \dots, L - 1$, and $\mathcal{Q}$ is used for storing the resulting data samples. Alternatively, we can write $\mathbf{Q}(:, :, l)$ in terms of its columns as

$$\mathbf{Q}(:, :, l) = [\mathbf{q}(:, 0, l), \mathbf{q}(:, 1, l), \dots, \mathbf{q}(:, N_c - 1, l)], \quad (28)$$

for $l = 0, 1, \dots, L - 1$, and $\mathbf{q}(:, n_c, l)$ is the $(n_c + 1)$ -th column of $\mathbf{Q}(:, :, l)$ whose elements are expressed as

$$\mathbf{q}(:, n_c, l) = [q(0, n_c, l), q(1, n_c, l), \dots, q(N_s - 1, n_c, l)]^T, \quad (29)$$

for $l = 0, 1, \dots, L - 1$ and $n_c = 0, 1, \dots, N_c - 1$. Since the beat frequency in one chirp duration is used for estimating the range of targets, we can view $\mathbf{q}(:, n_c, l)$ as the range-frequency spectrum[†] at the $(n_c + 1)$ -th chirp from the $(l + 1)$ -th antenna in the ULA. Thus, we call the DFT operation in (27) as the range-DFT.

Next, we use $\mathcal{R}$ and perform another DFT operation on each row of $\mathbf{Q}(:, :, l)$ in (28) to utilize the changes in beat frequencies across multiple chirp durations (i.e., Doppler frequency [3]), that is

$$\mathbf{R}(:, :, l) = \mathbf{Q}(:, :, l) \mathbf{F}_{N_c}^T, \quad (30)$$

for $l = 0, 1, \dots, L - 1$. Expressing $\mathbf{R}(:, :, l)$ in terms of its rows yields

$$\mathbf{R}(:, :, l) = [\mathbf{r}(0, :, l), \mathbf{r}(1, :, l), \dots, \mathbf{r}(N_s - 1, :, l)]^T, \quad (31)$$

for $l = 0, 1, \dots, L - 1$, and $\mathbf{r}^T(n_s, :, l)$ is the $(n_s + 1)$ -th row of $\mathbf{R}(:, :, l)$. The elements of $\mathbf{r}(n_s, :, l)$, for $n_s = 0, 1, \dots, N_s - 1$ and $l = 0, 1, \dots, L - 1$, are expressed as

$$\mathbf{r}(n_s, :, l) = [r(n_s, 0, l), r(n_s, 1, l), \dots, r(n_s, N_c - 1, l)]^T. \quad (32)$$

We can perceive $\mathbf{r}(n_s, :, l)$ as the Doppler-frequency spectrum[‡] at the $(n_s + 1)$ -th range-frequency point from the $(l + 1)$ -th antenna in the ULA, and the DFT operation in (30) is called as the Doppler-DFT.

Remark 1: The matrix $\mathbf{R}(:, :, l)$ in (31) provides a 2D range-Doppler-frequency spectrum-map, associated with the

[†]The unit is in digital angular range-frequency $2\pi n_s/N_s$ (radian/sample).

[‡]The unit is in digital angular Doppler-frequency $2\pi n_c/N_c$ (radian/sample).

output of the $(l + 1)$ -th antenna in the ULA. This 2D spectrum-map can be utilized to simultaneously obtain range and velocity information of the targets as in [26] and [27], and it is depicted in the toy example of Fig. 2.

For obtaining AoA information, we need to process the data samples along the spatial dimension to exploit differential distance from targets to each antenna in the ULA. Therefore, we use another matrix slice of $\mathcal{R}$, i.e., $\mathbf{R}(n_s, :, :)$, and the DFT operation is applied on each row of $\mathbf{R}(n_s, :, :)$ as

$$\mathbf{S}(n_s, :, :) = \mathbf{R}(n_s, :, :) \mathbf{F}_L^T, \quad (33)$$

for $n_s = 0, 1, \dots, N_s - 1$, and the resulting data samples are stored in $\mathcal{S}$. Representing $\mathbf{S}(n_s, :, :)$ by its rows gives

$$\mathbf{S}(n_s, :, :) = [\mathbf{s}(n_s, 0, :), \mathbf{s}(n_s, 1, :), \dots, \mathbf{s}(n_s, N_c - 1, :)]^T, \quad (34)$$

for $n_s = 0, 1, \dots, N_s - 1$, and $\mathbf{s}^T(n_s, n_c, :)$ is the $(n_c + 1)$ -th row of $\mathbf{S}(n_s, :, :)$. Furthermore, we can express the elements of $\mathbf{s}(n_s, n_c, :)$ as

$$\mathbf{s}(n_s, n_c, :) = [s(n_s, n_c, 0), s(n_s, n_c, 1), \dots, s(n_s, n_c, L - 1)]^T, \quad (35)$$

for $n_s = 0, 1, \dots, N_s - 1$ and $n_c = 0, 1, \dots, N_c - 1$. We interpret $\mathbf{s}(n_s, n_c, :)$ as the AoA spectrum at the $(n_s + 1)$ -th range-frequency and $(n_c + 1)$ -th Doppler-frequency points since it will be utilized for obtaining AoA information of the targets. Therefore, we call the DFT operation in (33) as the angle-DFT, and it is depicted in the toy example of $L = 4$ antennas in Fig. 3.

Remark 2: The tensor $\mathcal{S}$ provides a 3D range-Doppler-AoA spectrum-map, and we will discuss the units transformation in $\mathcal{S}$ for obtaining range, velocity, and AoA information of the targets in Section III-D.

D. RANGE, VELOCITY, AND AOA ESTIMATION

In this subsection, we present the estimation of range, velocity, and AoA based on the processed data samples in $\mathcal{S}$ from (33). Due to range-DFT and Doppler-DFT operations in Section III-C, the default units along n_s and n_c dimensions in $\mathcal{S}$ are digital angular range-frequencies and digital angular Doppler-frequencies, respectively. First, let us specify the grid of digital angular range-frequencies $\{\tilde{\Omega}_{R,n_s}\}_{n_s=0}^{N_s-1}$ and that of digital angular Doppler-frequencies $\{\tilde{\Omega}_{D,n_c}\}_{n_c=0}^{N_c-1}$ as

$$\tilde{\Omega}_{R,n_s} = 2\pi n_s/N_s, \quad n_s = 0, 1, \dots, N_s - 1,$$

and,

$$\tilde{\Omega}_{D,n_c} = 2\pi n_c/N_c, \quad n_c = 0, 1, \dots, N_c - 1,$$

respectively. Then, these two digital angular frequencies are converted to the grid of range and Doppler frequencies (with the units in Hertz) as follows

$$\tilde{f}_{R,n_s} = N_s \tilde{\Omega}_{R,n_s} / (2\pi T_c), \quad (36)$$

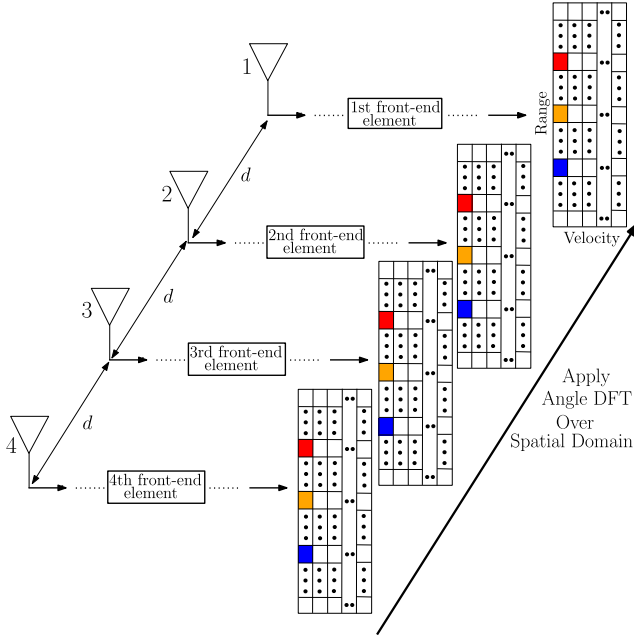


FIGURE 3. The ULA of $L = 4$ elements, and the corresponding 2D spectrum-maps produced after range-DFT and Doppler-DFT operations. Subsequently, the angle-DFT is performed over spatial domain to obtain the necessary data for AoA information.

and,

$$\tilde{f}_{D,n_c} = \tilde{\Omega}_{D,n_c} / (2\pi T_c), \quad (37)$$

for $n_s = 0, 1, \dots, N_s - 1$ and $n_c = 0, 1, \dots, N_c - 1$. Due to the normalized frequency in DFT operation, the conversion to Hertz in (36) and (37) involves multiplication with the sampling frequency N_s/T_c and the chirp rate $1/T_c$, respectively. For obtaining range and velocity information, we transform $\{\tilde{f}_{R,n_s}\}_{n_s=0}^{N_s-1}$ and $\{\tilde{f}_{D,n_c}\}_{n_c=0}^{N_c-1}$ into grid of ranges $\{\tilde{R}_{n_s}\}_{n_s=0}^{N_s-1}$ and velocities $\{\tilde{v}_{n_c}\}_{n_c=0}^{N_c-1}$, respectively. By using (21), $\tilde{R}_{n_s}$ and $\tilde{v}_{n_c}$ are given by

$$\tilde{R}_{n_s} = c\tilde{f}_{R,n_s} / (2\alpha), \quad n_s = 0, 1, \dots, N_s - 1, \quad (38)$$

and,

$$\tilde{v}_{n_c} = c\tilde{f}_{D,n_c} / (2f_c), \quad n_c = 0, 1, \dots, N_c - 1. \quad (39)$$

Remark 3: By considering (21), the grid of ranges $\tilde{R}_{n_s}$ in (38) includes the contribution coming from the initial distance of the target at the start of FMCW signal transmission and the displacement of the target during the FMCW transmission caused by the non-zero velocity of moving target.

Next, we transform the grid units along the l -dimension for acquiring the AoA information. First, we describe $\mathbf{F}_L^{-1}$ through its column vectors as

$$\mathbf{F}_L^{-1} = [\mathbf{f}_0, \mathbf{f}_1, \dots, \mathbf{f}_{L-1}],$$

where $\mathbf{f}_l$ is the $(l+1)$ -th column of $\mathbf{F}_L^{-1}$, and its elements are given by

$$\mathbf{f}_l = \frac{1}{L} [1, e^{j2\pi l/L}, e^{j2\pi 2l/L}, \dots, e^{j2\pi (L-1)l/L}]^T. \quad (40)$$

By comparing (40) and the expression for the antenna response $a(\theta_p)$ in (11), it is straightforward to deduce that $L\mathbf{f}_l$ can be considered as the array response vector at the direction of $\hat{\theta}_l$ given by

$$\hat{\theta}_l = \begin{cases} \sin^{-1}(2l/L), & \text{for } l = 0, 1, \dots, \lfloor L/2 \rfloor \\ \sin^{-1}(\frac{2l-2L}{L}), & \text{for } l = \lfloor \frac{L}{2} \rfloor + 1, \lfloor \frac{L}{2} \rfloor + 2, \\ & \dots, L-1 \end{cases} \quad (41)$$

where $\lfloor L/2 \rfloor$ produces the largest integer less than $L/2$. In other words, $L\mathbf{f}_l$ is equivalent to the array response vector at the grid of angles when $\Delta = 0.5$ wavelength, i.e., $\mathbf{a}(\hat{\theta}_l)$ for $l = 0, 1, \dots, L-1$.

Now, we proceed to obtain the 3D spectrum map containing range, velocity, and AoA information by computing the magnitude of the spectrum in $\mathcal{S}$, i.e., $|s(n_s, n_c, l)|$, for $n_s = 0, 1, \dots, N_s - 1$, $n_c = 0, 1, \dots, N_c - 1$, and $l = 0, 1, \dots, L-1$. Then, the three-dimensional units are transformed into grid of ranges $\{\tilde{R}_{n_s}\}_{n_s=0}^{N_s-1}$, grid of velocities $\{\tilde{v}_{n_c}\}_{n_c=0}^{N_c-1}$, and grid of AoA $\{\hat{\theta}_l\}_{l=0}^{L-1}$ by using (38), (39), and (41), respectively. The locations of the P highest peaks in the magnitude spectrum-map are regarded as the estimates of the range, velocity, and the AoA of the P targets. The overall FMCW radar signal processing to estimate range, velocity, and AoA discussed throughout Section III is summarized in Fig. 4.

Remark 4: This paper considers P targets where they are significantly less than the number of grid points, i.e., $P < L$, $P \ll N_c$, and $P \ll N_s$, and this condition implies that the tensor $\mathcal{S}$ contains mostly zero or insignificant values[§] except at the location of P grid points. In other words, $\mathcal{S}$ possesses a sparsity feature, and we will exploit this feature in Section IV to incorporate the CS theory aimed at reducing the data samples in $\mathcal{S}$ while maintaining the estimation capability and accuracy.

Lastly, it is worth mentioning that a precision issue in the estimation results may arise due to the actual range, velocity, and AoA having continuous values. In particular, when the gap between adjacent grid points is significantly larger than the resolution of FMCW radar mentioned in Section II-B, high estimation errors will occur more often because the grid points cannot facilitate precision readings on the range, velocity, and AoA values. Since the spectrum given by $\mathcal{S}$ is two-sided, the gap between adjacent grid points for range- and Doppler-frequency domain are given by

$$\Delta_{\tilde{R}} \approx 2\tilde{R}_{N_s-1}/N_s - 1, \quad (42)$$

and

$$\Delta_{\tilde{v}} \approx 2\tilde{v}_{N_c-1}/N_c - 1, \quad (43)$$

respectively. We note that the gap between adjacent grid points for the AoA-frequency domain is not uniform across

[§]These negligible values come from the presence of noise in the FMCW system.

$L - 1$ points, and it can be calculated exhaustively by using (41). To mitigate the issue when the actual range, velocity, and AoA of the targets are not precisely located on the specified grid of ranges $\{\tilde{R}_{n_s}\}_{n_s=0}^{N_s-1}$, velocities $\{\tilde{v}_{n_c}\}_{n_c=0}^{N_c-1}$, and AoA $\{\tilde{\theta}_l\}_{l=0}^{L-1}$, extra grid points can be included in addition to the existing grid of ranges, velocities, and azimuth angles. Additional grid points are obtained by using DFT matrices with a larger size in the range-DFT, Doppler-DFT, and AoA-DFT operations in (27), (30), and (33). In the range-DFT operation, we put additional zero rows after the last row of matrix slice $\mathbf{X}(:, :, l)$ in (27), for $l = 0, 1, \dots, L - 1$, before applying the range-DFT operation. In the Doppler-DFT operation, additional zero columns are padded after the last column of matrix slice $\mathbf{Q}(:, :, l)$ in (30), for $l = 0, 1, \dots, L - 1$, and it is followed by applying DFT operation on each row of $\mathbf{Q}(:, :, l)$. The zero-padding procedure is especially desirable for AoA-DFT operation since the number of antenna in the ULA (L) is usually limited, and these antennas directly influence the angular grid points. We add extra zero columns after the last column of matrix slice $\mathbf{R}(n_s, :, :)$ in (33), then DFT operation is performed on each row of $\mathbf{R}(n_s, :, :)$, for $n_s = 0, 1, \dots, N_s - 1$. It is important to note that the zero-padding procedure only increases the number of grid points in the frequency domain, and it does not increase the resolution of the estimated parameters, i.e., range, velocity, and AoA of the targets.

IV. COMPRESSIVE FMCW SIGNAL PROCESSING

In this section, we propose a scheme that integrates the CS theory and the FMCW signal processing for estimating range, velocity, and AoA information of the targets by utilizing the sparsity feature in $\mathcal{S}$. Our proposed scheme, called compressive FMCW signal processing, consists of two main blocks: 1) compression block, and 2) reconstruction block, as shown in Fig. 5. First, the compression block performs RF front-end processing, compression, and sampling onto the received FMCW signals for producing the compressed version of $\mathcal{X}$ (also called compressive measurements). Then, the recovery of $\mathcal{S}$ is carried out by the reconstruction block from the compressive measurements, and we justify the reconstruction process through a theoretical analysis.

A. SPATIAL, TEMPORAL, AND CHIRP COMPRESSIONS

In our proposed compressive FMCW signal processing, we perform the compression process in the order of spatial domain, time domain, and chirp domain. The compression is performed by incorporating a $m \times n$ sensing matrix $\mathbf{C}$ (with $m < n$), which transforms a signal vector into a smaller-dimension signal vector. In this paper, we do not describe the procedures for designing $\mathbf{C}$ in detail, and interested readers are referred to [38]. First, let us consider the vector of received FMCW signals $\mathbf{x}(t)$ from the L antennas in

the ULA, that is

$$\mathbf{x}(t) = [x_0(t), x_1(t), \dots, x_{L-1}(t)]^T,$$

where $x_l(t)$ is the output of the $(l + 1)$ -th antenna in the ULA from (11) for $l = 0, 1, \dots, L - 1$. Then, the compressed signal vector in spatial domain $\mathbf{z}(t)$ is obtained by

$$\mathbf{z}(t) = \mathbf{C}_s \mathbf{x}(t) = [z_0(t), z_1(t), \dots, z_{M_l-1}(t)]^T \quad (44)$$

where $\mathbf{C}_s$ is a $M_l \times L$ spatial compression matrix, and it is expressed as

$$\mathbf{C}_s = [\mathbf{c}_{s,0}, \mathbf{c}_{s,1}, \dots, \mathbf{c}_{s,L-1}], \quad (45)$$

where $\mathbf{c}_{s,l} = [c_{s,l}[0], c_{s,l}[1], \dots, c_{s,l}[M_l - 1]]^T$. The spatial compression leads to a smaller number of RF front-end processing (i.e., mixing and filtering operation) since $M_l < L$, which reduces implementation cost on the FMCW receiver systems.

Similar to the RF front-end processing presented in III-A, the mixing process is now performed on $z_{m_l}(t)$, and it is expressed as

$$\begin{aligned} \tilde{z}_{m_l}(t) &= z_{m_l}(t)s(t) = \sum_{l=0}^{L-1} c_{s,l}[m_l]x_l(t)s(t) \\ &= \sum_{l=0}^{L-1} c_{s,l}[m_l]\tilde{x}_l(t). \end{aligned} \quad (46)$$

Subsequently, the filtering operation is given by

$$\begin{aligned} z_{\text{IF},m_l}(t) &= \tilde{z}_{m_l}(t) \star h(t) = \sum_{l=0}^{L-1} c_{s,l}[m_l]\tilde{x}_l(t) \star h(t) \\ &= \sum_{l=0}^{L-1} c_{s,l}[m_l]x_{\text{IF},l}(t), \end{aligned} \quad (47)$$

where $z_{\text{IF},m_l}(t)$ is the compressed IF signal at the output of $(m_l + 1)$ -th RF front-end. Further compression processes for $z_{\text{IF},m_l}(t)$, i.e., time and chirp compression, is carried out by the analog-to-information converter (AIC) [39], [40], [41] as shown in Fig. 5. In Section IV-B, we will describe the reconstruction processes that use the compressive measurements produced by the AIC to recover $\mathcal{S}$ mentioned in Section III-C.

Now, we present a mathematical model for the compression processes in a discrete setup which will facilitate our theoretical analysis for the reconstruction processes in Section IV-B. Let us consider the time-domain samples of $z_{\text{IF},m_l}(t)$ at the $(n_c + 1)$ -th chirp duration, i.e. $z_{\text{IF},m_l}[n_s, n_c]$, and the vector consisting of $z_{\text{IF},m_l}[n_s, n_c]$ for $m_l = 0, 1, \dots, M_l - 1$ is given by

$$\begin{aligned} \mathbf{z}_{\text{IF}}[n_s, n_c] &= [z_{\text{IF},0}[n_s, n_c], z_{\text{IF},1}[n_s, n_c], \dots, \\ &\quad z_{\text{IF},M_l-1}[n_s, n_c]]^T, \end{aligned} \quad (48)$$

for $n_c = 0, 1, \dots, N_c - 1$ and $n_s = 0, 1, \dots, N_s - 1$. By noting (44), we alternatively express $\mathbf{z}_{\text{IF}}[n_s, n_c]$ as

$$\mathbf{z}_{\text{IF}}[n_s, n_c] = \mathbf{C}_s \mathbf{x}_{\text{IF}}[n_s, n_c], \quad (49)$$

[¶]It is also called a compression matrix.

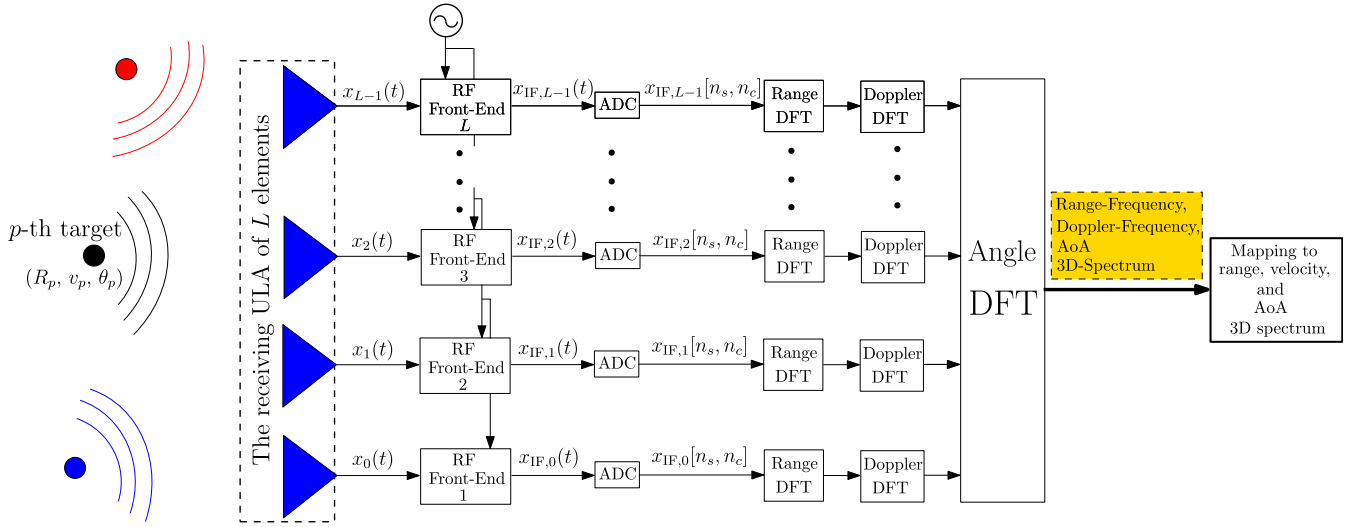


FIGURE 4. The conventional FMCW radar signal processing model to obtain range, velocity, and AoA information of the targets [32], [34], [36].

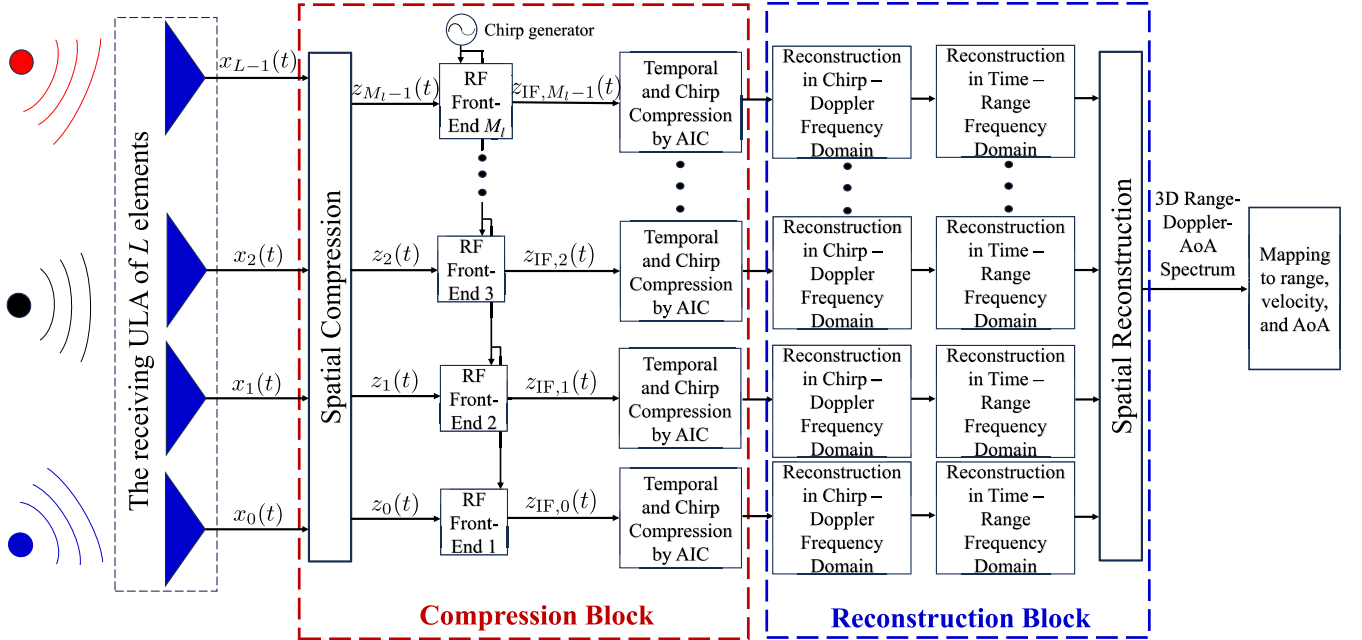


FIGURE 5. The proposed compressive FMCW signal processing for estimating range, velocity, and AoA of the targets. The received FMCW signal is first processed by the compression block, then the reconstruction block outputs the 3D range-Doppler-AoA spectrum.

since $\mathbf{x}_{IF}[n_s, n_c]$ is the collection of L discretized IF signals defined in (23). It is important to note that $x_{IF,l}[n_s, n_c]$ from (23) is the Nyquist-sampled version of $x_{IF,l}(t)$ at the $(n_c + 1)$ -th chirp duration, and our proposed compressive FMCW signal processing does not actually produce $x_{IF,l}[n_s, n_c]$ since the AIC unifies the sampling and compression processes [39]. Expressing (44) and the RF front-end processing in the discrete setup, i.e., (49), simplifies the mathematical model for the compression process.

Similar to (25), we express (49) in a $M_l \times N_s$ matrix $\mathbf{Z}_{IF}[n_c]$, i.e.,

$$\mathbf{Z}_{IF}[n_c] = \mathbf{C}_s \mathbf{X}_{IF}[n_c], \quad (50)$$

with $n_c = 0, 1, \dots, N_c - 1$, and its $(m_l + 1)$ -th row is filled with

$$[z_{IF,m_l}[0, n_c], z_{IF,m_l}[1, n_c], \dots, z_{IF,m_l}[N_s - 1, n_c]].$$

For processing data samples via the tensor format as in subsection IV-B, we express (50) as

$$\mathbf{Z}_s(:, n_c, :) = \mathbf{X}(:, n_c, :) \mathbf{C}_s^T, \quad (51)$$

where $\mathbf{Z}_s \in \mathbb{C}^{N_s \times N_c \times M_l}$, and $\mathbf{X}(:, n_c, :)$ is defined in (26). Next, we perform the compression in time domain, namely temporal compression, which is given by

$$\mathbf{Z}_t(:, n_c, :) = \mathbf{C}_t \mathbf{Z}_s(:, n_c, :), \quad (52)$$

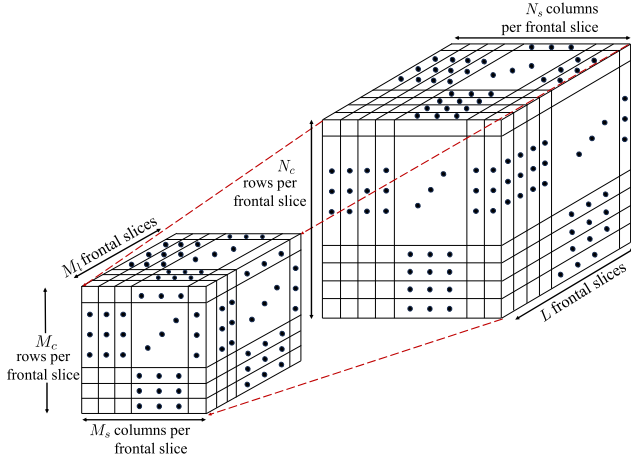


FIGURE 6. The reduced tensor containing data samples as a result of spatial-domain, time-domain, and chirp-domain compressions.

for $n_c = 0, 1, \dots, N_c - 1$, $\mathbf{Z}_t \in \mathbb{C}^{M_s \times N_c \times M_l}$, and $\mathbf{C}_t$ is a $M_s \times N_s$ temporal compression matrix. Lastly, the compression in chirp domain is given by

$$\mathbf{Z}_c(:, :, m_l) = \mathbf{Z}_t(:, :, m_l) \mathbf{C}_c^T, \quad (53)$$

for $m_l = 0, 1, \dots, M_l - 1$, $\mathbf{Z}_c \in \mathbb{C}^{M_s \times M_c \times M_l}$, and $\mathbf{C}_c$ is a $M_c \times N_c$ chirp compression matrix. The resulting tensor $\mathbf{Z}_c$ is the compressed version of $\mathcal{X}$ from Section III-C, and Fig. 6 illustrates the smaller number of data samples required for estimating range, velocity, and AoA information of the targets.

Lastly, it is important to note that the imperfections typically occur in the compression block [42], and we model it with the addition of noise as depicted in Fig. 7. As a result, the compression block outputs noisy compressive measurements $\hat{\mathbf{Z}}_c \in \mathbb{C}^{M_s \times M_c \times M_l}$, instead of $\mathbf{Z}_c$ in (53). Specifically, we have

$$\hat{\mathbf{Z}}_c(:, :, m_l) = \mathbf{Z}_c(:, :, m_l) + \mathbf{W}(:, :, m_l), \quad (54)$$

for $m_l = 0, 1, \dots, M_l - 1$, and the entries of $\mathbf{W} \in \mathbb{C}^{M_s \times M_c \times M_l}$ are taken from Gaussian random variables with zero-mean and variance σ^2 .

B. RECONSTRUCTION PROCESSES

In this subsection, we present techniques to reconstruct $\mathcal{S}$ in subsection III-C from the noisy compressive measurements stored in $\hat{\mathbf{Z}}_c$. The crux of reconstructing $\mathcal{S}$ lies in its sparsity feature, and we formalize this feature through Definition 1 and Definition 2.

Definition 1 (*P*-sparsity): A tensor, a matrix, or a vector possesses a *P*-sparsity if the tensor, the matrix, or the vector has at most *P* significant-valued entries and the remaining entries are zeros or negligible values. The tensor, matrix, and vector possessing a *P*-sparsity are called as *P*-sparse tensor, *P*-sparse matrix, and *P*-sparse vector, respectively.

Definition 2 (Jointly *P*-sparse vectors): Let the collection of vectors $\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_J$ for $J \in \mathbb{Z}^+$ be the columns of matrix

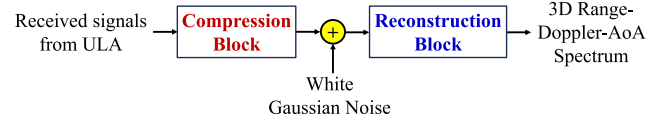


FIGURE 7. The addition of noise in the compression block.

$\mathbf{G}$, i.e., $\mathbf{G} = [\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_J]$. Then, the collection of such vectors in $\mathbf{G}$ is said to be jointly *P*-sparse vectors, when each of the vectors possesses a *P*-sparsity and $\mathbf{G}$ contains at most *P* rows having significant-valued entries.

Based on Definition 1 and Definition 2, we highlight the useful properties that are used extensively in this subsection in Lemma 1 and Lemma 2.

Lemma 1: The tensor $\mathcal{S}$ mentioned in Section III-C possesses a *P*-sparsity. Thus, all of its matrix slices, i.e., $\mathbf{S}(n_s, :, :)$ for $n_s = 0, 1, \dots, N_s - 1$, $\mathbf{S}(:, n_c, :)$ for $n_c = 0, 1, \dots, N_c - 1$, and $\mathbf{S}(:, :, l)$ for $l = 0, 1, \dots, L - 1$ also have *P*-sparsity. Furthermore, all $N_s \times 1$, $N_c \times 1$, and $L \times 1$ ($1 \times N_s$, $1 \times N_c$, and $1 \times L$, resp.) vectors contained in $\mathcal{S}$ are jointly *P*-sparse vectors.

Proof: Remark 4 implies that $\mathcal{S}$ possesses a *P*-sparsity. Consequently, each of its matrix slices has at most *P* significant-valued entries, which suggests that the number of rows having significant-valued entries in each of the matrix slices cannot exceed *P*. Hence, each of the matrix slices is *P*-sparse matrix. Now, let us consider a matrix $\tilde{\mathbf{S}}$ formed by a concatenation of matrix slices $\mathbf{S}(:, :, l)$ for $l = 0, 1, \dots, L - 1$, i.e.,

$$\tilde{\mathbf{S}} = [\mathbf{S}(:, :, 0), \mathbf{S}(:, :, 1), \dots, \mathbf{S}(:, :, L - 1)],$$

whose columns can be viewed as a collection of $N_s \times 1$ vectors in $\mathcal{S}$. Note that $\tilde{\mathbf{S}}$ can alternatively be constructed by concatenating $\mathbf{S}(n_s, :, :)$ for $n_s = 0, 1, \dots, N_s - 1$, and $\mathbf{S}^T(n_s, :, :)$ for $n_s = 0, 1, \dots, N_s - 1$, which will yield a collection of $N_c \times 1$ and $L \times 1$ vectors, respectively. Since $\mathcal{S}$ is *P*-sparse tensor, $\tilde{\mathbf{S}}$ has at most *P* rows having significant-valued entries. Therefore, all $N_s \times 1$, $N_c \times 1$, and $L \times 1$ ($1 \times N_s$, $1 \times N_c$, and $1 \times L$, resp.) vectors contained in $\mathcal{S}$ are jointly *P*-sparse vectors. $\square$

Lemma 2: A collection of vectors produced by linear combinations of jointly *P*-sparse vectors always consists of jointly *P*-sparse vectors.

Proof: Let us consider matrix $\tilde{\mathbf{G}}$ which is defined as

$$\tilde{\mathbf{G}} = [\tilde{\mathbf{g}}_1, \tilde{\mathbf{g}}_2, \dots, \tilde{\mathbf{g}}_J] = \mathbf{G}\mathbf{H}, \quad (55)$$

where $\tilde{\mathbf{g}}_1, \tilde{\mathbf{g}}_2, \dots, \tilde{\mathbf{g}}_J$ for $J \in \mathbb{Z}^+$ are the columns of $\tilde{\mathbf{G}}$, $\mathbf{G}$ is specified in Definition 2, and $\mathbf{H}$ is an arbitrary $I \times J$ matrix. By noting (55), it is straightforward to observe that $\tilde{\mathbf{g}}_1, \tilde{\mathbf{g}}_2, \dots, \tilde{\mathbf{g}}_J$ are the vectors produced by linear combinations of columns in $\mathbf{G}$, and these vectors belong to the column space of $\mathbf{G}$. Hence, $\tilde{\mathbf{g}}_1, \tilde{\mathbf{g}}_2, \dots, \tilde{\mathbf{g}}_J$ are jointly *P*-sparse vectors. $\square$

Now, we present the three reconstruction processes, and these processes attempt to reverse the compression processes

in Section IV-A which are necessary to obtain $\mathcal{S}$. However, a direct recovery of $\mathcal{Z}_t$, $\mathcal{Z}_s$, and $\mathcal{X}$ is not possible due to the non-invertible compression matrices used in (53), (52), and (51). As an alternative, we propose to initially reconstruct the frequency domain of $\mathcal{Z}_t$, $\mathcal{Z}_s$, and $\mathcal{X}$ by utilizing the sparsity feature in each of the frequency domains. Then, the IDFT operation is performed to recover $\mathcal{Z}_t$, $\mathcal{Z}_s$, and $\mathcal{X}$ after a successful reconstruction. To justify our approach, we present the theoretical results through Theorem 1, Theorem 2, and Theorem 3 which show the sparsity feature in the spectrum.

Remark 5: In the proof of Theorem 1, Theorem 2, and Theorem 3, we assume that the addition of noise in the compression block does not obstruct the P highest values in $\mathcal{S}$, such that Lemma 1 still holds.

In the first reconstruction process, we present the recovery of $\mathcal{Z}_t$ in (53) from $\hat{\mathcal{Z}}_c$. In Theorem 1, we show that the Doppler-frequency domain of each row in $\mathbf{Z}_t(:, :, m_l)$ possesses the P -sparsity feature, and the reconstruction of $\mathbf{Z}_t(:, :, m_l)$ for $m_l = 0, 1, \dots, M_l - 1$ follows afterwards.

Theorem 1: Let $\mathbf{Z}_{t,f}(:, :, m_l)$ for $m_l = 0, 1, \dots, M_l - 1$ be the matrix slice of the tensor $\mathcal{Z}_{t,f} \in \mathbb{C}^{M_s \times N_c \times M_l}$. Furthermore, let $\mathbf{z}_{t,f}^T(m_s, :, m_l)$ for $m_s = 0, 1, \dots, M_s - 1$ be the row vectors of $\mathbf{Z}_{t,f}(:, :, m_l)$, and these row vectors are Doppler-frequency domain of rows in $\mathbf{Z}_t(:, :, m_l)$. In other words, $\mathbf{Z}_{t,f}(:, :, m_l)$ can be expressed as

$$\mathbf{Z}_{t,f}(:, :, m_l) = [\mathbf{z}_{t,f}(0, :, m_l), \mathbf{z}_{t,f}(1, :, m_l), \dots, \mathbf{z}_{t,f}(M_s - 1, :, m_l)]^T = \mathbf{Z}_t(:, :, m_l) \mathbf{F}_{N_c}^T, \quad (56)$$

for $m_l = 0, 1, \dots, M_l - 1$. Then, $\mathbf{z}_{t,f}^T(m_s, :, m_l)$ for $m_s = 0, 1, \dots, M_s - 1$ and $m_l = 0, 1, \dots, M_l - 1$ contained in $\mathcal{Z}_{t,f}$ are jointly P -sparse vectors.

Proof: By utilizing (33), Lemma 1, and Lemma 2, the tensor $\mathcal{R}$ can be regarded as a collection of $N_c \times 1$ vectors, i.e.,

$$\mathbf{R}(n_s, :, :) = [\mathbf{r}(n_s, :, 0), \mathbf{r}(n_s, :, 1), \dots, \mathbf{r}(n_s, :, L - 1)],$$

for $n_s = 0, 1, \dots, N_s - 1$, which consists of jointly P -sparse vectors. In other words, the rows of $\mathbf{R}(:, :, l)$, i.e.,

$$\mathbf{R}(:, :, l) = [\mathbf{r}(0, :, l), \mathbf{r}(1, :, l), \dots, \mathbf{r}(N_s - 1, :, l)]^T,$$

for $l = 0, 1, \dots, L - 1$, are also jointly P -sparse vectors. Applying IDFT operation on each column of $\mathbf{R}(:, :, l)$, for $l = 0, 1, \dots, L - 1$, leads to

$$\dot{\mathbf{R}}(:, :, l) = \mathbf{F}_{N_s}^{-1} \mathbf{R}(:, :, l), \quad (57)$$

where $\dot{\mathbf{R}} \in \mathbb{C}^{N_s \times N_c \times L}$. Due to (57) and Lemma 2, $\dot{\mathbf{R}}$ contains a collection of $N_c \times 1$ vectors, i.e.,

$$\dot{\mathbf{R}}(:, :, l) = [\dot{\mathbf{r}}(0, :, l), \dot{\mathbf{r}}(1, :, l), \dots, \dot{\mathbf{r}}(N_s - 1, :, l)]^T,$$

for $l = 0, 1, \dots, L - 1$, which comprises of jointly P -sparse vectors. Subsequently, we perform a temporal compression on $\mathbf{R}(:, :, l)$, for $l = 0, 1, \dots, L - 1$, i.e.,

$$\ddot{\mathbf{R}}(:, :, l) = \mathbf{C}_t \dot{\mathbf{R}}(:, :, l), \quad (58)$$

where $\ddot{\mathbf{R}} \in \mathbb{C}^{M_s \times N_c \times L}$. Consequently, (58) and Lemma 2 implies that $\ddot{\mathbf{R}}$ contains $N_c \times 1$ vectors, i.e.,

$$\ddot{\mathbf{R}}(:, :, l) = [\ddot{\mathbf{r}}(0, :, l), \ddot{\mathbf{r}}(1, :, l), \dots, \ddot{\mathbf{r}}(M_s - 1, :, l)]^T,$$

for $l = 0, 1, \dots, L - 1$, which are jointly P -sparse vectors. Clearly, the columns of $\ddot{\mathbf{R}}(m_s, :, :)$ for $m_s = 0, 1, \dots, M_s - 1$, i.e.,

$$\ddot{\mathbf{R}}(m_s, :, :) = [\ddot{\mathbf{r}}(m_s, :, 0), \ddot{\mathbf{r}}(m_s, :, 1), \dots, \ddot{\mathbf{r}}(m_s, :, L - 1)],$$

are also jointly P -sparse vectors. Finally, we establish a relation between $\ddot{\mathbf{R}}(m_s, :, :)$ and $\mathbf{Z}_{t,f}(m_s, :, :)$ through

$$\mathbf{Z}_{t,f}(m_s, :, :) = \ddot{\mathbf{R}}(m_s, :, :) \mathbf{C}_s^T, \quad (59)$$

for $m_s = 0, 1, \dots, M_s - 1$. Alternatively, $\mathcal{Z}_{t,f}$ can be represented as a collection of $N_c \times 1$ column vectors $\mathbf{z}_{t,f}(m_s, :, m_l)$, i.e.,

$$\mathbf{Z}_{t,f}(m_s, :, :) = [\mathbf{z}_{t,f}(m_s, :, 0), \mathbf{z}_{t,f}(m_s, :, 1), \dots, \mathbf{z}_{t,f}(m_s, :, M_l - 1)], \quad (60)$$

for $m_s = 0, 1, \dots, M_s - 1$. By noting (59), (60), and Lemma 2, it is straightforward to deduce that $\mathbf{z}_{t,f}^T(m_s, :, m_l)$ for $m_s = 0, 1, \dots, M_s - 1$ and $m_l = 0, 1, \dots, M_l - 1$ contained in $\mathcal{Z}_{t,f}$ are jointly P -sparse vectors. $\square$

By using (53) and (56), $\mathbf{Z}_c(:, :, m_l)$ can be alternatively represented by

$$\mathbf{Z}_c(:, :, m_l) = \mathbf{Z}_{t,f}(:, :, m_l) (\mathbf{F}_{N_c}^{-1})^T \mathbf{C}_c^T, \quad (61)$$

for $m_l = 0, 1, \dots, M_l - 1$. For mathematical convenience, we use a slightly different view of (61), that is

$$\mathbf{Z}_c^T(:, :, m_l) = \mathbf{C}_c \mathbf{F}_{N_c}^{-1} \mathbf{Z}_{t,f}^T(:, :, m_l). \quad (62)$$

By considering (62), we establish the relation between a $M_c \times 1$ vector $\mathbf{z}_c(m_s, :, m_l)$ and a $N_c \times 1$ vector $\mathbf{z}_{t,f}(m_s, :, m_l)$ through

$$\mathbf{z}_c(m_s, :, m_l) = \Psi_c \mathbf{z}_{t,f}(m_s, :, m_l), \quad (63)$$

for $m_s = 0, 1, \dots, M_s - 1$, $m_l = 0, 1, \dots, M_l - 1$, and $\Psi_c = \mathbf{C}_c \mathbf{F}_{N_c}^{-1}$ has a size of $M_c \times N_c$. We observe that the system of equations in (63) is an underdetermined system since $M_c < N_c$, which implies that the solution $\mathbf{z}_{t,f}(m_s, :, m_l)$ is not unique for a given $\mathbf{z}_c(m_s, :, m_l)$. However, Theorem 1 enables us to reconstruct $\mathbf{z}_{t,f}(m_s, :, m_l)$ from the multiple measurement vectors (MMV), i.e., $\mathbf{z}_c(m_s, :, m_l)$ for $m_l = 0, 1, \dots, M_l - 1$, $m_s = 0, 1, \dots, M_s - 1$, due to the sparsity feature of $\mathbf{z}_{t,f}(m_s, :, m_l)$. In addition, Ψ_c already satisfies the mutual coherence and RIP [20], [21] due to the design of $\mathbf{C}_c$ [38], and it facilitates a recovery of the

sparse solutions $\mathbf{z}_{t,f}(m_s, :, m_l)$ for $m_l = 0, 1, \dots, M_l - 1$ and $m_s = 0, 1, \dots, M_s - 1$.

Well-known algorithms have been proposed to obtain sparse solutions in the underdetermined system with MMV, such as MMV Basic Matching Pursuit (M-BMP) [42], [43], MMV Orthogonal Matching Pursuit (M-OMP) [44], MMV Order Recursive Matching Pursuit (M-ORMP) [42], [45], and MMV Focal Underdetermined System Solver (M-FOCUSS) [42]. Since this work assumes the presence of additive white Gaussian noise (AWGN) in the compression block, we consider Regularized M-FOCUSS (RM-FOCUSS) algorithm [42] for reconstructing $\mathbf{z}_{t,f}(m_s, :, m_l)$ from the noisy MMV.[†] In particular, we specify $\hat{\mathbf{z}}_c(m_s, :, m_l)$ for $m_s = 0, 1, \dots, M_s - 1$ and $m_l = 0, 1, \dots, M_l - 1$ as the noisy MMV, i.e.,

$$\hat{\mathbf{z}}_c(m_s, :, m_l) = \mathbf{z}_c(m_s, :, m_l) + \mathbf{w}(m_s, :, m_l), \quad (64)$$

where $\mathbf{w}(m_s, :, m_l)$ is a $M_c \times 1$ noise vector from (54), and $\hat{\mathbf{z}}_c(m_s, :, m_l)$ is taken from the $(m_s + 1)$ -th column of $\hat{\mathbf{Z}}_c^T(:, :, m_l)$ in (54). Due to the presence of noise in (64), the RM-FOCUSS algorithm generates $\hat{\mathbf{z}}_{t,f}(m_s, :, m_l)$, i.e., the estimate of $\mathbf{z}_{t,f}(m_s, :, m_l)$ in (63).

After obtaining the estimates $\hat{\mathbf{z}}_{t,f}(m_s, :, m_l)$ for $m_s = 0, 1, \dots, M_s - 1$ and $m_l = 0, 1, \dots, M_l - 1$ from the RM-FOCUSS algorithm, we use the tensor $\hat{\mathbf{Z}}_{t,f} \in \mathbb{C}^{M_s \times N_c \times M_l}$ for storing $\hat{\mathbf{z}}_{t,f}(m_s, :, m_l)$. Specifically, we construct

$$\hat{\mathbf{Z}}_{t,f}(:, :, m_l) = [\hat{\mathbf{z}}_{t,f}(0, :, m_l), \hat{\mathbf{z}}_{t,f}(1, :, m_l), \dots, \hat{\mathbf{z}}_{t,f}(M_s - 1, :, m_l)]^T, \quad (65)$$

for $m_l = 0, 1, \dots, M_l - 1$. By using (65), the estimate of $\mathbf{Z}_t$ in (53), i.e., $\hat{\mathbf{Z}}_t \in \mathbb{C}^{M_s \times N_c \times M_l}$, is readily obtained by

$$\hat{\mathbf{Z}}_t(:, :, m_l) = \hat{\mathbf{Z}}_{t,f}(:, :, m_l)(\mathbf{F}_{N_c}^T)^{-1}, \quad (66)$$

for $m_l = 0, 1, \dots, M_l - 1$.

Now, we describe the second reconstruction process which aims to recover $\mathbf{Z}_s$ in (52) from $\hat{\mathbf{Z}}_t$. For a mathematical convenience in reconstructing $\mathbf{Z}_s$, we note the equivalent expression of (52) which is given by

$$\mathbf{Z}_t(:, :, m_l) = \mathbf{C}_t \mathbf{Z}_s(:, :, m_l), \quad (67)$$

for $m_l = 0, 1, \dots, M_l - 1$. Theorem 2 shows that the range-frequency domain of each column in $\mathbf{Z}_s(:, :, m_l)$ has the P -sparsity feature, then we present the reconstruction of $\mathbf{Z}_s(:, :, m_l)$ for $m_l = 0, 1, \dots, M_l - 1$.

Theorem 2: Let $\mathbf{Z}_{s,f}(:, :, m_l)$ for $m_l = 0, 1, \dots, M_l - 1$ be the matrix slice of the tensor $\mathbf{Z}_{s,f} \in \mathbb{C}^{N_s \times N_c \times M_l}$. Furthermore, let $\mathbf{z}_{s,f}(:, n_c, m_l)$ for $n_c = 0, 1, \dots, N_c - 1$ be the column vectors of $\mathbf{Z}_{s,f}(:, :, m_l)$, and these column vectors are range-frequency domain of columns in $\mathbf{Z}_s(:, :, m_l)$. In other words, $\mathbf{Z}_{s,f}(:, :, m_l)$ can be expressed as

$$\mathbf{Z}_{s,f}(:, :, m_l) = [\mathbf{z}_{s,f}(:, 0, m_l), \mathbf{z}_{s,f}(:, 1, m_l), \dots, \mathbf{z}_{s,f}(:, N_c - 1, m_l)] = \mathbf{F}_{N_s} \mathbf{Z}_s(:, :, m_l), \quad (68)$$

[†]In this paper, we do not mention the procedures of RM-FOCUSS algorithm, and interested readers are referred to [42].

for $m_l = 0, 1, \dots, M_l - 1$. Then, $\mathbf{z}_{s,f}(:, n_c, m_l)$ for $n_c = 0, 1, \dots, N_c - 1$ and $m_l = 0, 1, \dots, M_l - 1$ contained in $\mathbf{Z}_{s,f}$ are jointly P -sparse vectors.

Proof: Instead of using (33), we note that $\mathcal{S}$ can be alternatively constructed through

$$\mathbf{S}(:, n_c, :) = \mathbf{R}(:, n_c, :)\mathbf{F}_L^T, \quad (69)$$

for $n_c = 0, 1, \dots, N_c - 1$. Due to (69), Lemma 1, and Lemma 2, the tensor $\mathcal{R}$ can be viewed as a collection of $N_s \times 1$ vectors, i.e.,

$$\mathbf{R}(:, n_c, :) = [\mathbf{r}(:, n_c, 0), \mathbf{r}(:, n_c, 1), \dots, \mathbf{r}(:, n_c, L - 1)],$$

for $n_c = 0, 1, \dots, N_c - 1$, which are jointly P -sparse vectors. In other words, the columns of $\mathbf{R}(:, :, l)$, i.e.,

$$\mathbf{R}(:, :, l) = [\mathbf{r}(:, 0, l), \mathbf{r}(:, 1, l), \dots, \mathbf{r}(:, N_c - 1, l)],$$

for $l = 0, 1, \dots, L - 1$, are also jointly P -sparse vectors. Therefore, by noting (30) and Lemma 2, the tensor $\mathcal{Q}$ consists of $N_s \times 1$ vectors, i.e.,

$$\mathbf{Q}(:, :, l) = [\mathbf{q}(:, 0, l), \mathbf{q}(:, 1, l), \dots, \mathbf{q}(:, N_c - 1, l)]$$

for $l = 0, 1, \dots, L - 1$, which are jointly P -sparse vectors. Hence, the columns of $\mathbf{Q}(:, n_c, :)$, i.e.,

$$\mathbf{Q}(:, n_c, :) = [\mathbf{q}(:, n_c, 0), \mathbf{q}(:, n_c, 1), \dots, \mathbf{q}(:, n_c, L - 1)],$$

for $n_c = 0, 1, \dots, N_c - 1$, are also jointly P -sparse vectors. Now, we establish the relation between $\mathbf{Q}(:, n_c, :)$ and $\mathbf{Z}_{s,f}(:, n_c, :)$ through

$$\mathbf{Z}_{s,f}(:, n_c, :) = \mathbf{Q}(:, n_c, :)\mathbf{C}_s^T, \quad (70)$$

for $n_c = 0, 1, \dots, N_c - 1$. Alternatively, $\mathbf{Z}_{s,f}$ can be represented as a collection of $N_s \times 1$ vectors, i.e.,

$$\mathbf{Z}_{s,f}(:, n_c, :) = [\mathbf{z}_{s,f}(:, n_c, 0), \mathbf{z}_{s,f}(:, n_c, 1), \dots, \mathbf{z}_{s,f}(:, n_c, M_l - 1)], \quad (71)$$

for $n_c = 0, 1, \dots, N_c - 1$. By using (70), (71), and Lemma 2, we conclude that $\mathbf{z}_{s,f}(:, n_c, m_l)$ for $n_c = 0, 1, \dots, N_c - 1$ and $m_l = 0, 1, \dots, M_l - 1$ contained in $\mathbf{Z}_{s,f}$ are jointly P -sparse vectors. $\square$

By using (67) and (68), $\mathbf{Z}_t(:, :, m_l)$ can be alternatively expressed as

$$\mathbf{Z}_t(:, :, m_l) = \mathbf{C}_t \mathbf{F}_{N_s}^{-1} \mathbf{Z}_{s,f}(:, :, m_l), \quad (72)$$

for $m_l = 0, 1, \dots, M_l - 1$. Then, we establish the relation between a $M_s \times 1$ vector $\mathbf{z}_t(:, n_c, m_l)$ and a $N_s \times 1$ vector $\mathbf{z}_{s,f}(:, n_c, m_l)$ by using (72), that is

$$\mathbf{z}_t(:, n_c, m_l) = \Psi_t \mathbf{z}_{s,f}(:, n_c, m_l), \quad (73)$$

for $m_l = 0, 1, \dots, M_l - 1$, $n_c = 0, 1, \dots, N_c - 1$, and $\Psi_t = \mathbf{C}_t \mathbf{F}_{N_s}^{-1}$. Theorem 2 allows us to reconstruct $\mathbf{z}_{s,f}(:, n_c, m_l)$ for

$m_l = 0, 1, \dots, M_l - 1, n_c = 0, 1, \dots, N_c - 1$ by using the RM-FOCUSS algorithm. However, we note that $\mathbf{z}_t(:, n_c, m_l)$ is not available for the reconstruction process, and $\hat{\mathbf{z}}_t(:, n_c, m_l)$ taken from $\hat{\mathbf{Z}}_t(:, :, m_l)$ in (66) is used instead.

After obtaining the estimates of $\mathbf{z}_{s,f}(:, n_c, m_l)$, i.e., $\hat{\mathbf{z}}_{s,f}(:, n_c, m_l)$ for $n_c = 0, 1, \dots, N_c - 1$ and $m_l = 0, 1, \dots, M_l - 1$ from the RM-FOCUSS algorithm, we use a tensor $\hat{\mathbf{Z}}_{s,f} \in \mathbb{C}^{N_s \times N_c \times M_l}$ for storing $\hat{\mathbf{z}}_{s,f}(:, n_c, m_l)$, i.e.,

$$\hat{\mathbf{Z}}_{s,f}(:, :, m_l) = [\hat{\mathbf{z}}_{s,f}(:, 0, m_l), \hat{\mathbf{z}}_{s,f}(:, 1, m_l), \dots, \hat{\mathbf{z}}_{s,f}(:, N_c - 1, m_l)], \quad (74)$$

for $m_l = 0, 1, \dots, M_l - 1$. Then, the estimate of $\mathbf{Z}_s$, i.e., $\hat{\mathbf{Z}}_s \in \mathbb{C}^{N_s \times N_c \times M_l}$, is obtained by

$$\hat{\mathbf{Z}}_s(:, :, m_l) = \mathbf{F}_{N_s}^{-1} \hat{\mathbf{Z}}_{s,f}(:, :, m_l), \quad (75)$$

for $m_l = 0, 1, \dots, M_l - 1$.

As the third reconstruction process, we aim to recover the AoA domain of $\mathcal{X}$ in (51), and Theorem 3 shows that the AoA domain of each row in $\mathbf{X}(:, n_c, :)$ possess the P -sparsity feature.

Theorem 3: Let $\mathbf{X}_f(:, n_c, :)$ for $n_c = 0, 1, \dots, N_c - 1$ be the matrix slice of the tensor $\mathcal{X}_f \in \mathbb{C}^{N_s \times N_c \times L}$. Furthermore, let $\mathbf{x}_f^T(n_s, n_c, :)$ for $n_s = 0, 1, \dots, N_s - 1$ be the row vectors of $\mathbf{X}_f(:, n_c, :)$, and these row vectors are AoA domain of rows in $\mathbf{X}(:, n_c, :)$. In other words, $\mathbf{X}_f(:, n_c, :)$ can be expressed as

$$\mathbf{X}_f(:, n_c, :) = [\mathbf{x}_f(0, n_c, :), \mathbf{x}_f(1, n_c, :), \dots, \mathbf{x}_f(N_s - 1, n_c, :)]^T = \mathbf{X}(:, n_c, :)\mathbf{F}_L^T, \quad (76)$$

for $n_c = 0, 1, \dots, N_c - 1$. Then, $\mathbf{x}_f^T(n_s, n_c, :)$ for $n_s = 0, 1, \dots, N_s - 1$ and $n_c = 0, 1, \dots, N_c - 1$ contained in $\mathcal{X}_f$ are jointly P -sparse vectors.

Proof: First, we introduce a tensor $\dot{\mathcal{S}} \in \mathbb{C}^{N_s \times N_c \times L}$, which is constructed through

$$\dot{\mathcal{S}}(:, n_c, :) = \mathbf{F}_{N_s}^{-1} \mathbf{S}(:, n_c, :), \quad (77)$$

for $n_c = 0, 1, \dots, N_c - 1$. Clearly, by noting (77), Lemma 1 and Lemma 2, $\dot{\mathcal{S}}$ consists of $1 \times L$ vectors, i.e.,

$$\dot{\mathcal{S}}(:, n_c, :) = [\dot{\mathbf{s}}(0, n_c, :), \dot{\mathbf{s}}(1, n_c, :), \dots, \dot{\mathbf{s}}(N_s - 1, n_c, :)]^T,$$

for $n_c = 0, 1, \dots, N_c - 1$, which are jointly P -sparse vectors. In other words, the rows of $\dot{\mathcal{S}}(n_s, :, :)$, i.e.,

$$\dot{\mathcal{S}}(n_s, :, :) = [\dot{\mathbf{s}}(n_s, 0, :), \dot{\mathbf{s}}(n_s, 1, :), \dots, \dot{\mathbf{s}}(n_s, N_c - 1, :)]^T,$$

for $n_s = 0, 1, \dots, N_s - 1$, are also jointly P -sparse vectors. Now, we describe a relation between $\dot{\mathcal{S}}(n_s, :, :)$ and $\mathbf{X}_f(n_s, :, :)$ for $n_s = 0, 1, \dots, N_s - 1$, through

$$\dot{\mathcal{S}}(n_s, :, :) = \mathbf{F}_{N_c} \mathbf{X}_f(n_s, :, :). \quad (78)$$

By viewing $\mathcal{X}_f$ as a collection of $1 \times L$ vectors, we have

$$\mathbf{X}_f(n_s, :, :) = [\mathbf{x}_f(n_s, 0, :), \mathbf{x}_f(n_s, 1, :), \dots, \mathbf{x}_f(n_s, N_c - 1, :)]^T, \quad (79)$$

for $n_s = 0, 1, \dots, N_s - 1$. By considering (78), (79), and Lemma 2, we conclude that $\mathbf{x}_f^T(n_s, n_c, :)$ for $n_s = 0, 1, \dots, N_s - 1$ and $n_c = 0, 1, \dots, N_c - 1$ contained in $\mathcal{X}_f$ are jointly P -sparse vectors. $\square$

By using (51) and (76), $\mathbf{Z}_s(:, n_c, :)$ can be alternatively expressed as

$$\mathbf{Z}_s(:, n_c, :) = \mathbf{X}_f(:, n_c, :)(\mathbf{F}_L^T)^{-1} \mathbf{C}_s^T, \quad (80)$$

for $n_c = 0, 1, \dots, N_c - 1$. For mathematical convenience, we present a slightly different view of (80), that is

$$\mathbf{Z}_s^T(:, n_c, :) = \mathbf{C}_s \mathbf{F}_L^{-1} \mathbf{X}_f^T(:, n_c, :). \quad (81)$$

By using (81), we establish the relation between a $M_l \times 1$ vector $\mathbf{z}_s(n_s, n_c, :)$ and a $L \times 1$ vector $\mathbf{x}_f(n_s, n_c, :)$, that is

$$\mathbf{z}_s(n_s, n_c, :) = \Psi_s \mathbf{x}_f(n_s, n_c, :), \quad (82)$$

for $n_s = 0, 1, \dots, N_s - 1, n_c = 0, 1, \dots, N_c - 1$, and $\Psi_s = \mathbf{C}_s \mathbf{F}_L^{-1}$ is an $M_l \times L$ matrix. Theorem 3 justifies the reconstruction of $\mathbf{x}_f(n_s, n_c, :)$ from $\mathbf{z}_s(n_s, n_c, :)$ by using the RM-FOCUSS algorithm. Since $\mathbf{z}_s(n_s, n_c, :)$ is not available, we use $\hat{\mathbf{z}}_s(n_s, n_c, :)$ taken from $\hat{\mathbf{Z}}_s(:, :, m_l)$ in (75) for reconstructing $\mathbf{x}_f(n_s, n_c, :)$.

After the estimates of $\mathbf{x}_f(n_s, n_c, :)$, i.e., $\hat{\mathbf{x}}_f(n_s, n_c, :)$ for $n_s = 0, 1, \dots, N_s - 1$ and $n_c = 0, 1, \dots, N_c - 1$ are obtained from the RM-FOCUSS algorithm, we use a tensor $\hat{\mathcal{X}}_f \in \mathbb{C}^{N_s \times N_c \times L}$ for storing $\hat{\mathbf{x}}_f(n_s, n_c, :)$. In other words, we construct

$$\hat{\mathcal{X}}_f(:, n_c, :) = [\hat{\mathbf{x}}_f(0, n_c, :), \hat{\mathbf{x}}_f(1, n_c, :), \dots, \hat{\mathbf{x}}_f(N_s - 1, n_c, :)]^T, \quad (83)$$

for $n_c = 0, 1, \dots, N_c - 1$. Subsequently, the estimate of $\mathcal{X}$ in (51), i.e., $\hat{\mathcal{X}} \in \mathbb{C}^{N_s \times N_c \times L}$, is readily obtained as

$$\hat{\mathcal{X}}(:, n_c, :) = \hat{\mathcal{X}}_f(:, n_c, :)(\mathbf{F}_L^T)^{-1}, \quad (84)$$

for $n_c = 0, 1, \dots, N_c - 1$.

Since the reconstruction process in this subsection aims to recover $\mathcal{S}$, we can avoid the operation in (84) by observing that the rows of $\hat{\mathcal{X}}_f(:, n_c, :)$ in (83) are already in the AoA domain. Therefore, $\hat{\mathcal{X}}_f$ is used directly to obtain

$$\hat{\mathcal{S}}(:, :, l) = \mathbf{F}_{N_s} \hat{\mathcal{X}}_f(:, :, l) \mathbf{F}_{N_c}^T, \quad (85)$$

for $l = 0, 1, \dots, L - 1$, and $\hat{\mathcal{S}} \in \mathbb{C}^{N_s \times N_c \times L}$. In (85), we use the notation $\hat{\mathcal{S}}$, i.e., the estimate of $\mathcal{S}$, since the addition of noise in the compression block leads to $\hat{\mathcal{S}} \neq \mathcal{S}$. Finally, the 3D spectrum-map containing range, velocity, and AoA information is obtained by processing $\hat{\mathcal{S}}$ with units transformation steps presented in Section III-D as shown in Fig. 5.

Considering the precision issue mentioned in Section III-D, we propose to increase the number of grid points in the AoA domain since L (i.e., the number of antennas) controls the number of grid points in the AoA domain, and it is often desirable to avoid large number of antennas in the ULA for cost-efficiency. To this end, we modify Ψ_s in (82) by employing a $L \times \hat{L}$ matrix $\hat{\mathbf{F}}$, i.e.,

$$\hat{\mathbf{F}} = \mathbf{F}_L^{-1},$$

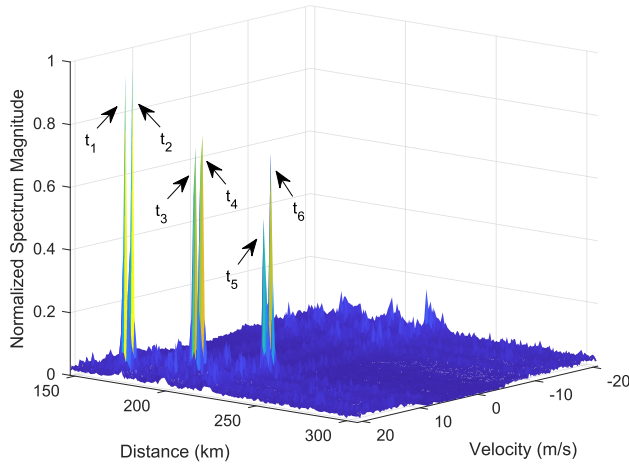


FIGURE 8. The normalized 2D magnitude-spectrum in the range-Doppler-frequency domain for Scenario 1, obtained by taking summation on $\hat{\mathcal{S}}$ over $\hat{L}$ -dimension.

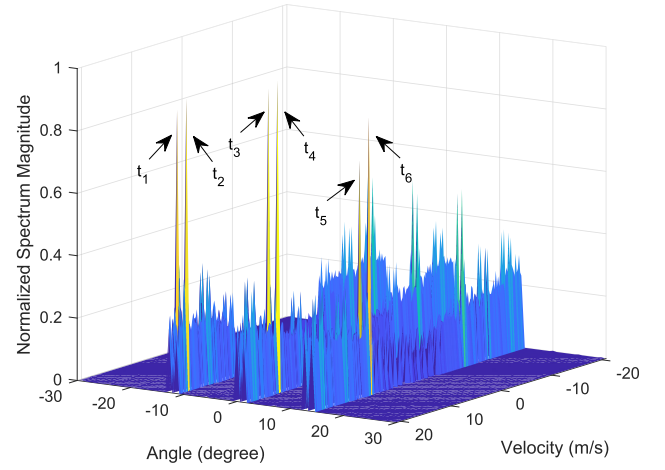


FIGURE 10. The normalized 2D magnitude-spectrum in the Doppler-frequency-AoA domain for Scenario 1, obtained by taking summation on $\hat{\mathcal{S}}$ over n_s -dimension.

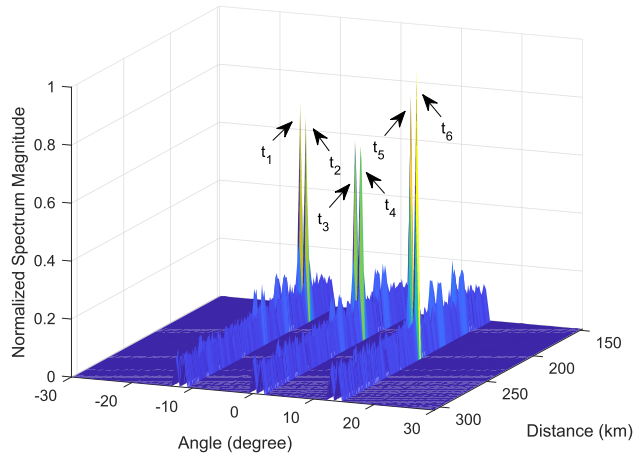


FIGURE 9. The normalized 2D magnitude-spectrum in the range-frequency-AoA domain for Scenario 1, obtained by taking summation on $\hat{\mathcal{S}}$ over n_c -dimension.

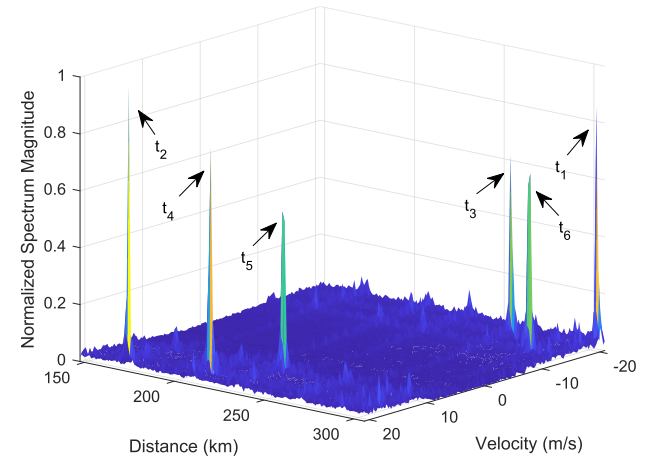


FIGURE 11. The normalized 2D magnitude-spectrum in the range-Doppler-frequency domain for Scenario 2, obtained by taking summation on $\hat{\mathcal{S}}$ over $\hat{L}$ -dimension.

where $\hat{\mathbf{I}}$ is obtained by removing the last $\hat{L} - L$ rows of identity matrix $\mathbf{I}_{\hat{L}}$. Therefore, (82) becomes

$$\mathbf{z}_s(n_s, n_c, :) = \hat{\Psi}_s \hat{\mathbf{x}}_f(n_s, n_c, :), \quad (86)$$

where $\hat{\Psi}_s = \mathbf{C}_s \hat{\mathbf{F}}$. Consequently, the RM-FOCUSS algorithm outputs $\hat{L}$ -dimensional sparse solutions $\hat{\mathbf{x}}_f(n_s, n_c, :)$ for $n_s = 0, 1, \dots, N_s - 1$ and $n_c = 0, 1, \dots, N_c - 1$. Similar to (85), we obtain the 3D spectrum-map with extra grid points in the AoA domain, i.e., $\hat{\mathcal{S}} \in \mathbb{C}^{N_s \times N_c \times \hat{L}}$ through

$$\hat{\mathcal{S}}(:, :, \hat{l}) = \mathbf{F}_{N_s} \hat{\mathbf{x}}_f(:, :, \hat{l}) \mathbf{F}_{N_c}^T, \quad (87)$$

for $\hat{l} = 0, 1, \dots, \hat{L} - 1$, and $\hat{\mathbf{x}}_f \in \mathbb{C}^{N_s \times N_c \times \hat{L}}$ is filled with $\hat{\mathbf{x}}_f(n_s, n_c, :)$ in similar fashion to (83).

V. SIMULATION RESULTS

In this section, the numerical simulations are conducted to show the capability and accuracy of FMCW radar employing

our proposed compressive signal processing in estimating range, velocity, and AoA of the targets. Particularly, we first show that the proposed compressive FMCW signal processing maintains the estimation capability and sensitivity, while using significantly fewer digital samples than conventional FMCW signal processing. Then, we demonstrate the accuracy of estimated range, velocity, and AoA from our proposed scheme by the numerical computation of root mean squared error (RMSE) [46]. Throughout this section, the simulations are carried out with the operating frequency of the FMCW system $f_c = 18$ MHz, which follows the typical operating frequency used in a popular FMCW radar application, i.e., HFSWR [9]. The other simulation settings for the FMCW system are determined by setting the maximum estimated range $R_{\max}$ and velocity $v_{\max}$ to around 300 km and 20 m/s, respectively. The values of $R_{\max}$ and $v_{\max}$ are chosen to facilitate HFSWR applications, which

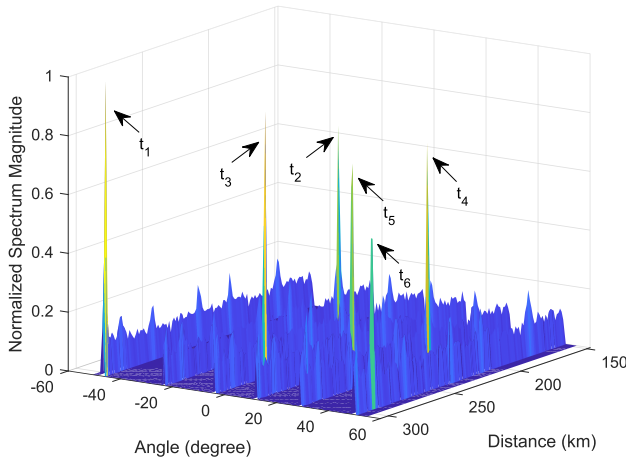


FIGURE 12. The normalized 2D magnitude-spectrum in the range-frequency-AoA domain for Scenario 2, obtained by taking summation on $\hat{S}$ over n_c -dimension.

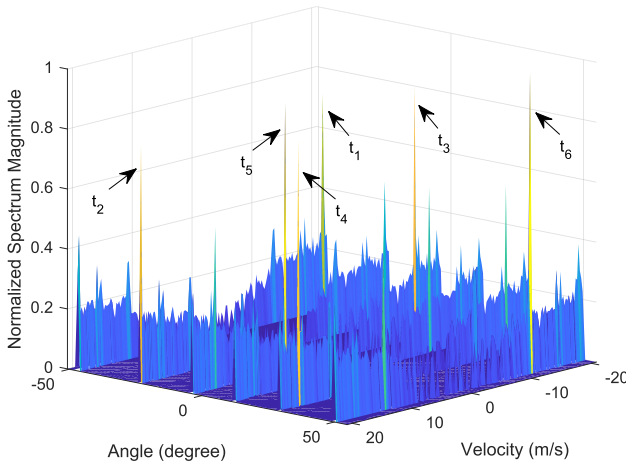


FIGURE 13. The normalized 2D magnitude-spectrum in the Doppler-frequency-AoA domain for Scenario 2, obtained by taking summation on $\hat{S}$ over n_s -dimension.

involve detecting moving vessels from the coastal baseline. Additionally, the RM-FOCUSS parameters are heuristically adjusted following the suggestion of [42]. The complete simulation settings used in this section are summarized in Table 1.

To show the estimation capability of our proposed scheme, we consider Scenario 1 and Scenario 2, in which the FMCW radar transmits a frame of chirps to estimate $P = 6$ targets with parameters specified in Table 2 and 3, respectively. These scenarios demonstrate challenging estimation conditions, where targets in Scenario 1 are closely located with almost similar velocity, and Scenario 2 has several targets whose actual parameters are near $R_{\max}$ and $v_{\max}$ of the FMCW radar. Then, by using the received signals at the ULA, we employ the compressive signal processing presented in Section IV to produce the estimation of range, velocity, and AoA of the targets. The compression rates

TABLE 1. The setup for compressive FMCW signal processing throughout Section V.

Parameters	Value
Carrier frequency (f_c)	18 MHz
Wavelength (λ)	16.67 meters
Chirp duration (T_c)	0.2 seconds
Chirp bandwidth (B_c)	125 kHz
Number of chirps in a single FMCW frame (N_c)	181 chirps
Number of samples in a single chirp (N_s)	512 samples
Maximum estimated range ($R_{\max}$) according to (9)	307.2 km
Maximum estimated velocity ($v_{\max}$) according to (10)	20.83 m/s
Gap between adjacent grid points in range estimates ($\Delta_{\hat{R}}$) according to (42)	1.2 km
Gap between adjacent grid points in velocity estimates ($\Delta_{\hat{v}}$) according to (43)	0.2315 m/s
Number of antennas in the ULA (L)	26 antennas
Distance between antennas in the ULA ($\lambda/2$)	8.33 meters
Amplitude of received chirps (V_y)	1 mV
Increased grid points in AoA domain ($\hat{L}$)	181
RM-FOCUSS convergence criterion	0.001
RM-FOCUSS diversity measure	0.8
RM-FOCUSS regularization measure	0.001

TABLE 2. The parameters of $P = 6$ targets for Scenario 1.

Target	Distance (km)	Velocity (m/s)	AoA (degree)
t_1	171.6	18.75	-12.7675
t_2	174	18.287	-11.4724
t_3	195.6	14.1204	0
t_4	198	13.6574	1.2663
t_5	219.6	9.4907	12.7675
t_6	222	9.0278	14.0692

TABLE 3. The parameters of $P = 6$ targets for Scenario 2. The moving targets with direction toward the FMCW receiver are denoted by negative velocities.

Target	Distance (km)	Velocity (m/s)	AoA (degree)
t_1	304.8	-20.3704	-46.8262
t_2	172.8	20.1389	-23.4401
t_3	268.8	-16.6667	-3.8014
t_4	208.8	17.1296	29.0903
t_5	232.8	11.8056	12.1191
t_6	292.8	-12.2685	49.679

of spatial domain (M_l/L), time domain (M_s/N_s), and chirp domain (M_c/N_c) for both scenarios are set to 0.8077, 0.5, and 0.4033, respectively.

Fig. 8, Fig. 9, and Fig. 10 depict the normalized 2D magnitude spectrum for Scenario 1 which are obtained by taking summation over $\hat{l}$ -, n_c -, and n_s -dimension in $\hat{S}$, respectively. Based on these figures, we summarize the estimation results of range, velocity, and AoA for Scenario 1 in Table 4, and these estimates clearly demonstrate the capability of our proposed scheme to resolve closely located targets with a minor velocity difference. Similarly,

TABLE 4. The estimated parameters of $P = 6$ targets for Scenario 1.

Target Index (p)	Distance (km)	Velocity (m/s)	AoA (degree)
t_1	172.8	19.213	-12.7675
t_2	175.2	18.75	-11.4724
t_3	196.8	14.5833	0
t_4	199.2	14.1204	1.2663
t_5	219.6	9.9537	12.7675
t_6	222	9.4907	14.0692

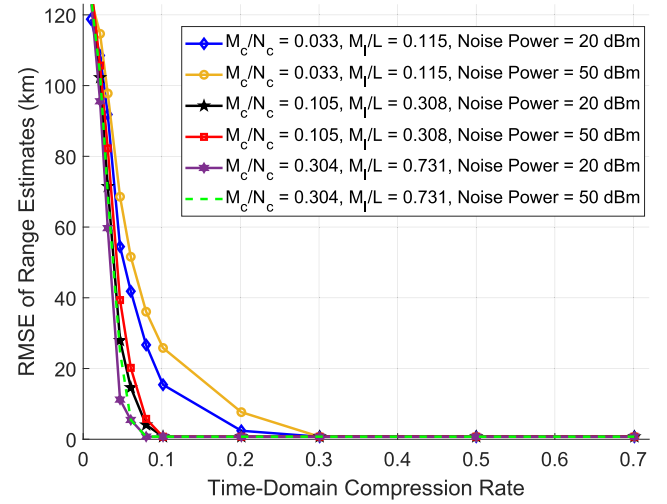
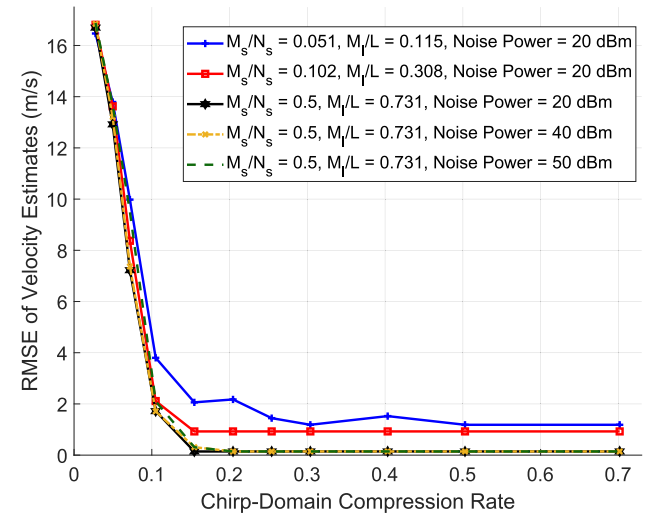
TABLE 5. The estimated parameters of $P = 6$ targets for Scenario 2. The moving targets with direction toward the FMCW receiver are denoted by negative velocities.

Target	Distance (km)	Velocity (m/s)	AoA (degree)
t_1	303.6	-20.1389	-46.8262
t_2	174	20.6019	-23.4401
t_3	267.6	-16.4352	-3.8014
t_4	210	17.5926	29.0903
t_5	232.8	12.2685	12.1191
t_6	292.8	-12.037	49.679

we visualize the normalized 2D magnitude spectrum for Scenario 2 in Fig. 11, Fig. 12, and Fig. 13, then Table 5 presents the resulting estimated parameters. The results from Table 5 show that our proposed compressive signal processing is capable of estimating range and velocity whose actual values are close to $R_{\max}$ and $v_{\max}$, respectively, while only using $\sim 16\%$ of the digital samples needed for conventional FMCW signal processing (i.e., $N_s N_c L = 2409472$ samples). It is worth mentioning that using lower sampling-rate in the conventional FMCW signal processing such that the number of samples matches to our proposed scheme ($M_s M_c M_l = 392448$ samples) will yield maximum range and velocity of 153.6 km and 8.4024 m/s, respectively, which are far below $R_{\max}$ and $v_{\max}$ from Table 1.

Now, we evaluate the estimation accuracy of our proposed compressive FMCW signal processing in a scenario where the actual range, velocity, and AoA of a target is randomized. Fig. 14 depicts the RMSE performance of range estimates with variation of M_s/N_s in $[0.0098, 0.7012]$ under different sets of M_c/N_c , M_l/L , and σ^2 . Clearly, the RMSE curve is a decreasing function of M_s/N_s as more compressed data samples lead to significant loss in range information. By using the strong compression set, i.e., $M_c/N_c = 0.033$ and $M_l/L = 0.115$, we observe a slightly better RMSE performance under $\sigma^2 = 20$ dBm compared to $\sigma^2 = 50$ dBm. The RMSE values reach a plateau at around 0.7 km for $M_s/N_s \geq 0.3$ regardless of the values of σ^2 , and these RMSE values indicate an excellent estimation accuracy since they are smaller than $\Delta_{\hat{R}}$. However, higher values of M_c/N_c and M_l/L enable us to achieve a RMSE value of 0.7 km at $M_s/N_s < 0.3$.

Subsequently, Fig. 15 shows the decreasing RMSE curve of velocity estimates for $M_c/N_c \in [0.0276, 0.7017]$ with varying sets of M_s/N_s , M_l/L , and σ^2 . When $M_s/N_s = 0.5$, $M_l/L = 0.731$, and $M_c/N_c \geq 0.15$, highly accurate estimates are observed since the RMSE values stay at around

**FIGURE 14.** The RMSE performance of the estimated range as a function of M_s/N_s with various sets of M_c/N_c , M_l/L and σ^2 .**FIGURE 15.** The RMSE performance of the estimated velocity as a function of M_c/N_c with various sets of M_s/N_s , M_l/L , and σ^2 .

0.14 m/s which are less than $\Delta_{\hat{v}}$ regardless the variation of σ^2 . In contrast, worse RMSE values are obtained for stronger compression sets, i.e., $M_s/N_s \leq 0.102$ and $M_l/L \leq 0.308$, which implies a significant loss in velocity information due to the compression process.

Lastly, Fig. 16 presents the decaying function of RMSE performance related to AoA estimates with $M_l/L \in [0.1538, 0.8077]$ under various sets of M_s/N_s , M_c/N_c , and σ^2 . When $\sigma^2 = 20$ dBm and $M_l/L \geq 0.3462$, the RMSE values of less than 0.34 degree is observed for all variation of M_s/N_s and M_c/N_c . When $\sigma^2 = 50$ dBm, a similar level of RMSE performance is achieved with $M_l/L \geq 0.4231$, $M_s/N_s \geq 0.051$, and $M_c/N_c \geq 0.105$. However, by using $M_s/N_s = 0.004$, and $M_c/N_c = 0.006$, the addition of noise with $\sigma^2 = 50$ dBm greatly distorts the estimation results which yields the RMSE values of greater than 18.46 degree with varying M_l/N_l . By observing the results

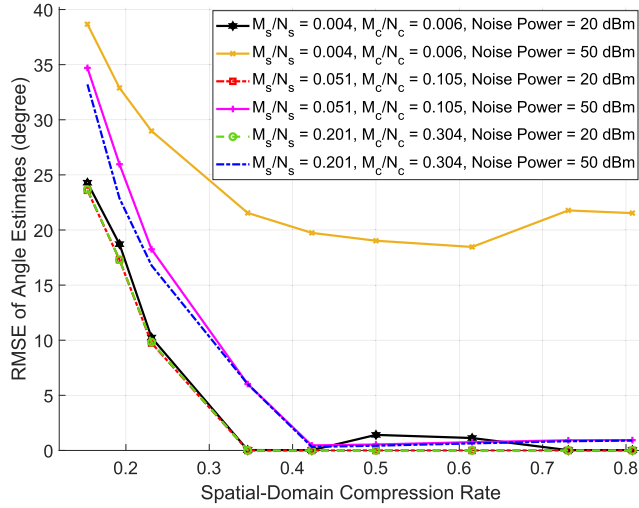


FIGURE 16. The RMSE performance of the estimated AoA as a function of M_1/L with various sets of M_s/N_s , M_c/N_c , and σ^2 .

of numerical simulations in Fig. 14, Fig. 15, and Fig. 16, we conclude that the proposed compressive FMCW signal processing achieves a highly accurate parameters estimation with properly adjusted compression rates in a specific noise level.

VI. CONCLUSION

This paper proposes a novel radar signal processing technique, namely the compressive FMCW signal processing, for estimating range, velocity, and AoA of the targets by utilizing the CS theory. The proposed technique aims to avoid using high-rate ADCs and decrease the number of RF front-end elements, thus reducing cost, power consumption, and number of digital samples in the FMCW signal processing systems without sacrificing the estimation capability and accuracy. To justify our CS-based approach, we first develop a new mathematical model for FMCW signal processing in a tensor format to facilitate a systematic and convenient analytical framework. Then, we show the sparsity feature of the 3D range-Doppler-AoA spectrum through an extensive theoretical analysis, which enables us to incorporate the CS theory into the FMCW signal processing. The simulation results show that our proposed scheme maintains the estimation capability and accuracy of the conventional FMCW signal processing, while using a significantly smaller number of digital samples. As a future research direction, various algorithms to recover sparse solutions with noisy MMV should be investigated. In addition, it is worth comparing the computational complexity of compressive FMCW signal processing with different sparse solutions recovery algorithms for ensuring an efficient and practical compressive signal processing strategy in FMCW radar applications.

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